

DM545/DM871
Linear and Integer Programming

Slides: Full Deck

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5. oktober 2026

Outline

1. Course Organization
2. Preliminaries
3. Introduction: Operations Research
 - Resource Allocation
 - Duality
4. Introduction
 - Diet Problem
5. Solving LP Problems
 - Fourier-Motzkin method
6. Mathematical Programming
 - Definitions
 - Fundamental Theorem of LP
 - Gaussian Elimination

7. Simplex Method
 - Standard Form
 - Basic Feasible Solutions
 - Algorithm
 - Tableaux and Dictionaries
8. Exception Handling
9. Initialization
10. Derivation and Motivation
11. Theory
12. Derivation
 - Lagrangian Duality
13. Dual Simplex
14. Sensitivity Analysis

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Who are we?

47 in total registered in ItsLearning

DM545 (5 ECTS)

33 enrolled

- Math-economics
(Bachelor, 5th semester)
Section M1
6 enrolled
- Others?
Section H1
27 enrolled

DM871 (5 ECTS)

13 enrolled

- Section H1, 13 enrolled
 - Computer Science
(Master)
 - Mathematics
(Bachelor/Master)
 - Applied Mathematics
(Bachelor/Master)
 - Data Science (Master)
 - Others?

Aims of the course

Learn about **mathematical optimization**:

- linear programming (continuous/numerical linear optimization)
- integer linear programming (discrete/combinatorial linear optimization)

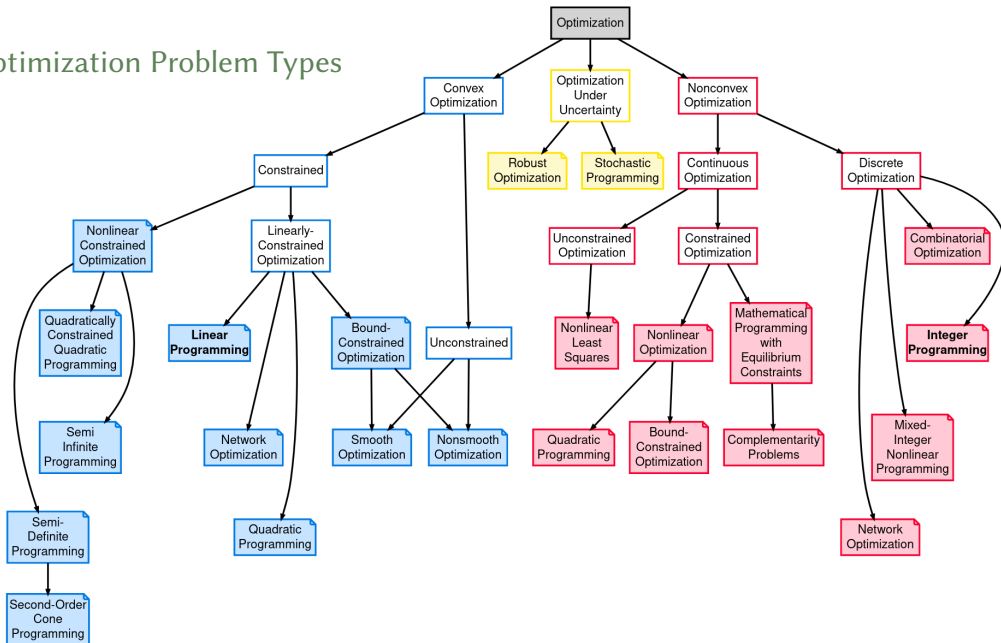
Hence, an alternative course title could be: **LINEAR OPTIMIZATION**

Optimization is an important tool in **decision making** and in analyzing physical systems.

In mathematical terms, an **optimization problem** is the problem of finding the **best solution** from the set of all **feasible solutions**.

The first step in the optimization process is constructing an appropriate **mathematical model**

Optimization Problem Types



Contents of the Course (aka Syllabus)

Linear Programming

- 1 Introduction - Linear Programming, Notation & Modeling
- 2 Linear Programming, Simplex Method
- 3 Exception Handling
- 4 Duality Theory
- 5 Sensitivity
- 6 Revised Simplex Method

Integer Linear Programming

- 7 Modeling Examples, Good Formulations, Relaxations
- 8 Well Solved Problems
- 9 Cutting Planes & Branch and Bound
- 10 Network Optimization Models (Max Flow, Min cost flow, Matching)
- 11 More on Modeling

Practical Information

Teacher: Marco Chiarandini (imada.sdu.dk/u/marco/)

Instructor: (Marco Chiarandini ?)

Schedule, alternative views:

- mitsdu.sdu.dk, SDU Mobile
- Official course description (læserplanen)
- ItsLearning

Schedule (15 weeks, up to week 51):

- Introductory classes: 28 hours (14 classes), scheduled 32
- Training classes: 14 hours (7 classes), scheduled 28

Communication Means

- **Announcements** in ItsLearning
- Ask peers and teacher in class
- ⇔ External Web Page
(link <https://dm871.github.io>)
- Write to Marco (marco@imada.sdu.dk) for confidential and individual issues
- Feedback from classes
- Midway evaluation

~> **It is good to ask questions!!**

~> **Let me know if you think we should do things differently!**

Sources — Reading Material

Main references:

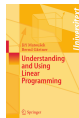
[LN] Lecture Notes (continuously updated)

[F] M. Fischetti, **Introduction to Mathematical Optimization**,
Independently published, 2019



Linear Programming:

[VA] R. Vanderbei. Linear Programming: Foundations and Extensions. Springer US, 2008



Integer Programming:

[Wo] L.A. Wolsey. Integer programming. John Wiley & Sons, New York, USA, 1998



Others

[HL] Frederick S Hillier and Gerald J Lieberman, Introduction to Operations Research, 9th edition, 2010

Course Material

External Web Page is the main reference for list of contents

It contains:

- List of units with topics and references
- Slides
- Exercises and some solutions
- Links
- Tutorials for programming tasks

Assessment

- One **3 h Written Exam**

January 11, 2027

all helping tools allowed, but not GenAI

- style: short open answers about calculations and modeling
 - (language: Danish and/or English)
- (only DM871) Home implementation assignment in the second half of the course

Graded with internal censorship

Training Sessions

- Do the plus exercises in advance
- Prepare the starred exercises in advance to get out the most from the session
- Participate actively in class
- Try the others, unsolved in class, after the session
- Best if carried out in small groups
- (Starred) Exercises are examples of exam questions

Exercise classes are about material presented the same week.

Prerequisites: Linear Algebra

Linear Algebra:

manipulation of matrices and vectors with some theoretical background

Linear Algebra

- Matrices and vectors - Matrix algebra

- Dot (scalar, Euclidean inner) product

- Geometric insights

- Systems of Linear Equations - Row echelon form, Gaussian elimination

- Matrix inversion and determinants

- Rank and linear dependency

Unit 0: sheet0.pdf

Prerequisites: Coding

- gives you the ability to create new and useful artifacts
- allows you to have more control of your world as more and more of it becomes digital
- is just fun.

It can also help you to **understand math**.

- listening to lectures
- watching an instructor work through a derivation
- working through numerical examples by hand
- learn **by doing, interacting with Python**.
- Python 3.14+, numpy
- ipython, jupyter, jupyterLab (= interactive python) or **Visual Code** or Spyder3.
- Commercial software: Gurobi, Cplex, Xpress $\approx$ 100 000 Dkk
Open source: Highs, SCIP
- Python APIs: gurobipy, Pyomo, Python-MIP, PySCIPOpt (Python interfaces to SCIP Optimization Suite)

A Note on Generative AI

- Read SDU rules on the topic <https://mitsdu.dk/en/aiatsdu>. They apply here.
In short: give credit when using
- Positive aspects:
better explanations, examples, immediate feedback
- Negative aspects:
economical costs, impact on environment, sovereignty risk, the social aspect, personal skills
- First learn, then use, and always check
Companies recommend to learn to use but want experts who can verify the outcomes

Use it responsibly, to enhance your learning!

Be aware that the exam form requires you to have achieved the learning goals not GenAI.

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Sets

- A **set** is a collection of objects. eg.: $A = \{1, 2, 3\}$
- $A = \{n \mid n \text{ is a whole number and } 1 \leq n \leq 3\}$
($\mid$ reads 'such that')
- $B = \{x \mid x \text{ is a student of this course}\}$
- $x \in A$
 x belongs to A
- set of no members: **empty set**, denoted $\emptyset$
- if a set S is a **(proper) subset** of a set T , we write $(S \subset T) \quad T \supseteq S$
 $\{1, 2, 5\} \subset \{1, 2, 4, 5, 6, 30\}$
- for two sets A and B , the **union** $A \cup B$ is $\{x \mid x \in A \text{ or } x \in B\}$
- for two sets A and B , the **intersection** $A \cap B$ is $\{x \mid x \in A \text{ and } x \in B\}$
 $A = \{1, 2, 3, 5\}$ and $B = \{2, 4, 5, 7\}$, then $A \cap B = \{2, 5\}$

Numbers

- set of real numbers: $\mathbb{R}$
- set of natural numbers: $\mathbb{N} = \{1, 2, 3, 4, \dots\}$ (positive integers); $\mathbb{N}_0$ to include zero
- set of all integers: $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$; $\mathbb{Z}_0^+$ only positives and zero
- set of rational numbers: $\mathbb{Q} = \{p/q \mid p, q \in \mathbb{Z}, q \neq 0\}$
- set of complex numbers: $\mathbb{C}$
- absolute value (non-negative):

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a \leq 0 \end{cases}$$

- the set $\mathbb{R}^2$ is the set of ordered pairs (x, y) of real numbers
(eg, coordinates of a point wrt a pair of axes, the **Cartesian plane**)

Matrices and Vectors

- A **matrix** is a rectangular array of numbers or symbols. It can be written as

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

- An $n \times 1$ matrix is a **column vector**, or simply a vector:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

- the set $\mathbb{R}^n$ is the set of vectors $[x_1, x_2, \dots, x_n]^T$ of real numbers (eg, coordinates of a point wrt an n -dimensional space, the **Euclidean Space**)

Graphs

- An **undirected graph** $G=(V,E)$ is a structure made of a pair of sets: V the set of **vertices** and $E \subseteq \{e = \{u, v\} \mid u, v \in V\}$ the set of **edges** (undirected links)
- A **directed graph (digraph)** $D = (V, A)$ is a pair of sets: V the set of **vertices** and $A \subseteq \{uv = (u, v) \mid u, v \in V\}$ the set of **arcs** (directed links)
- Graph drawing (2D embeddings)
- Labelled graphs and weighted graphs
- Walks, Paths, Cycles, Connectedness
- Problems on graphs: eg, determining intrinsic properties such as shortest *st*-path
- Matrix representations: **adjacency matrix**, **incidence matrix**

- adjacency set (neighborhood set) for $v \in V$: $N(v) = \{u \mid u \in V, uv \in E\}$; vertex degree (number of incident edges), $\delta(v)$; outdegree (number of outgoing edges), $\delta^+(v)$; indegree (number of incoming edges), $\delta^-(v)$;
- subgraph and **induced subgraph** $G' = G[V] \subseteq G$
- cut: set of edges that separates a connected graphs into two connected components $G[X], G[Y]$:
 $C = \{xy \mid x \in X, y \in Y\}$
- bipartite graphs $G = (U, V, E)$, trees (= connected acycle graphs)
- multi-graphs, hypergraphs, mixed graphs
- Graphs are the basic subject studied by graph theory. The word “graph” was first used in this sense by J.J. Sylvester in 1878 due to a direct relation between mathematics and chemical structure.

Basic Algebra

Elementary Algebra: the study of symbols and the rules for manipulating symbols. It differs from **arithmetic** in the use of abstractions, such as using letters to stand for numbers that are either unknown or allowed to take on many values

- collecting up terms: eg. $2a + 3b - a + 5b = a + 8b$
- multiplication of variables: eg:

$$a(-b) - 3ab + (-2a)(-4b) = -ab - 3ab + 8ab = 4ab$$

- expansion of bracketed terms: eg:

$$\begin{aligned} -(a - 2b) &= -a + 2b, \\ (2x - 3y)(x + 4y) &= 2x^2 - 3xy + 8xy - 12y^2 \\ &= 2x^2 + 5xy - 12y^2 \end{aligned}$$

- $a^r a^s = a^{r+s}$, $(a^r)^s = a^{rs}$, $a^{-n} = 1/a^n$,
 $a^{m/n} = (a^{1/n})^m$, $a^{1/n} = x \iff x^n = a$, $a^x = n \iff x = \log_a n$

Variables

- In Mathematics and Statistics, a **variable** is an alphabetic character representing a **value**, which is unknown. They are used in **symbolic** calculations.
Commonly given one-character names.
- in contrast, a **parameter** or **constant** or **given** is a known real number
- in contrast, in **Computer Science**, a **variable** is a storage location paired with an associated identifier, which contains a value that may be known or unknown.
Commonly given long, explanatory names.

Functions

- a **function** f on a set $\mathcal{X}$ into a set $\mathcal{Y}$ is a rule that assigns a **unique** element $f(x)$ in $\mathcal{Y}$ to each element x in $\mathcal{X}$.

$$y = f(x)$$

y dependent
variable

x independent
variable

- scalar (element of a field, eg, $\mathbb{R}$, vector of a single component) and vectors
- a **linear function** is a polynomial of degree one or less.
If only one variable:

$$f(x) := ax + b$$

where a and b are constants, often real numbers.

The graph is a nonvertical line (a is the slope and b the intercept).

For any finite number of variables: $f(x_1, \dots, x_k) = b + a_1 x_1 + \dots + a_k x_k$ and the graph is a hyperplane of dimension k . (actually these are affine functions: linear + translation)

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What is Operations Research?

Operations Research (aka, Management Science, Analytics):
is the discipline that uses a **scientific approach to decision making**.

It seeks to determine how best to design and operate a system,
usually under conditions requiring the allocation of scarce resources,
by means of **mathematics** and **computer science**.

Quantitative methods for planning and analysis.

It encompasses a wide range of problem-solving techniques and methods applied in the pursuit of improved decision-making and efficiency:

- simulation,
- **mathematical optimization**,
- queueing theory and other stochastic-process models,
- Markov decision processes
- econometric methods,
- data envelopment analysis,
- machine learning,
- expert systems

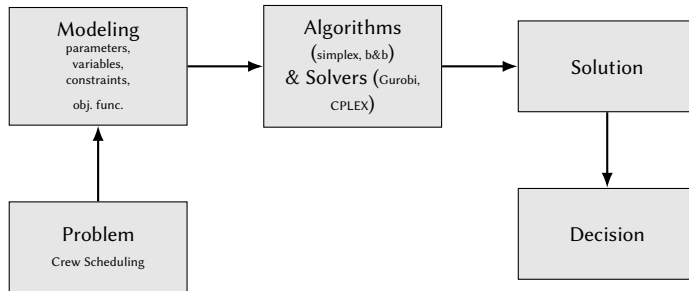
Some Examples ...

- Production Planning and Inventory Control
- Budget Investment
- Blending and Refining
- Manpower Planning
 - Crew Rostering (airline crew, rail crew, nurses)
- Packing Problems
 - Knapsack Problem
- Cutting Problems
 - Cutting Stock Problem
- Routing
 - Vehicle Routing Problem (trucks, planes, trains ...)
- Locational Decisions
 - Facility Location
- Scheduling/Timetabling
 - Examination timetabling/ train timetabling
- + many more

Common Characteristics

- Planning decisions must be made
- The problems relate to quantitative issues
 - Cheapest
 - Shortest route
 - Fewest number of people
- Not all plans are feasible - there are constraining rules
 - Limited amount of available resources
- It can be extremely difficult to figure out what to do

OR - The Process?



1. Observe the System
2. Formulate the Problem
3. Formulate Mathematical Model
4. Verify Model
5. Select Alternative
6. Show Results to Company
7. Implementation

Central Idea

Build a mathematical model describing exactly what one wants, and what the “rules of the game” are. However, **what is a mathematical model and how?**

Mathematical Modeling

- Find out exactly what the decision maker needs to know:
 - which investment?
 - which product mix?
 - which job j should a person i do?
- Define **Parameters**, that is the given, known elements of the problem.
- Define **Decision Variables** of suitable type (continuous, integer, binary) corresponding to the needs
- Formulate **Objective Function** computing the benefit/cost
- Formulate mathematical **Constraints** indicating the interplay between the different variables.

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Resource Allocation

In manufacturing industry, **factory planning**: find the best product mix.

Example

A factory makes two products **standard** and **deluxe**. Eg, yogurt, sleeping beds, etc.

A unit of **standard** gives a profit of 6(k) Dkk.

A unit of **deluxe** gives a profit of 8(k) Dkk.

The warming|grinding and cooling|polishing times in terms of hours per week for a **unit** of each type of product are given below:

	Standard	Deluxe
(Machine 1) Warming Grinding	5	10
(Machine 2) Cooling Polishing	4	4

Warming|Grinding capacity: 60 hours per week

Cooling|Polishing capacity: 40 hours per week

Q: How much of each product, **standard** and **deluxe**, should we produce to maximize the profit?

Mathematical Model

Decision Variables

$x_1 \geq 0$ units of product standard

$x_2 \geq 0$ units of product deluxe

Object Function

$\max 6x_1 + 8x_2$ maximize profit

Constraints

$5x_1 + 10x_2 \leq 60$ machine 1 capacity

$4x_1 + 4x_2 \leq 40$ machine 2 capacity

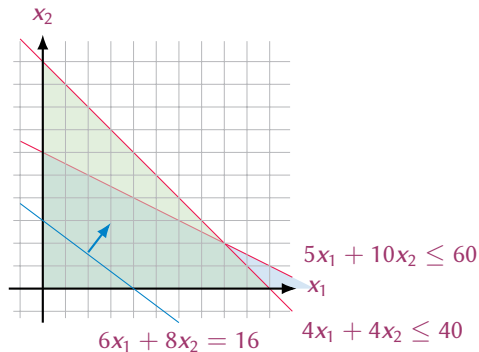
Mathematical Model

Machines/Materials A and B
Products 1 and 2

$$\begin{aligned}\max \quad & 6x_1 + 8x_2 \\ & 5x_1 + 10x_2 \leq 60 \\ & 4x_1 + 4x_2 \leq 40 \\ & x_1 \geq 0 \\ & x_2 \geq 0\end{aligned}$$

a_{ij}	1	2	b_i
A	5	10	60
B	4	4	40
c_j	6	8	

Graphical Representation:



Resource Allocation - General Model

Managing a production facility

$j = 1, 2, \dots, n$ products

$i = 1, 2, \dots, m$ materials

b_i units of raw material at disposal

a_{ij} units of raw material i to produce one unit of product j

σ_j market price of unit of j th product

ρ_i prevailing market value for material i

$c_j = \sigma_j - \sum_{i=1}^m \rho_i a_{ij}$ profit per unit of product j

x_j amount of product j to produce

$$\begin{array}{ll} \max & c_1 x_1 + c_2 x_2 + c_3 x_3 + \dots + c_n x_n = z \\ \text{subject to} & a_{11} x_1 + a_{12} x_2 + a_{13} x_3 + \dots + a_{1n} x_n \leq b_1 \\ & a_{21} x_1 + a_{22} x_2 + a_{23} x_3 + \dots + a_{2n} x_n \leq b_2 \\ & \dots \\ & a_{m1} x_1 + a_{m2} x_2 + a_{m3} x_3 + \dots + a_{mn} x_n \leq b_m \\ & x_1, x_2, \dots, x_n \geq 0 \end{array}$$

Notation

$$\begin{array}{ll}\max & c_1x_1 + c_2x_2 + c_3x_3 + \dots + c_nx_n = z \\ \text{s.t.} & a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n \leq b_1 \\ & a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n \leq b_2 \\ & \dots \\ & a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n \leq b_m \\ & x_1, x_2, \dots, x_n \geq 0\end{array}$$

$$\begin{array}{ll}\max & \sum_{j=1}^n c_jx_j \\ & \sum_{j=1}^n a_{ij}x_j \leq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \quad j = 1, \dots, n\end{array}$$

In Matrix Form

$$\begin{array}{ll}\max & c_1x_1 + c_2x_2 + c_3x_3 + \dots + c_nx_n = z \\ \text{s.t.} & a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n \leq b_1 \\ & a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n \leq b_2 \\ & \dots \\ & a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n \leq b_m \\ & x_1, x_2, \dots, x_n \geq 0\end{array}$$

$$\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\begin{array}{ll}\max & z = \mathbf{c}^T \mathbf{x} \\ & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq 0\end{array}$$

Our Numerical Example

$$\begin{aligned} \max \quad & \sum_{j=1}^n c_j x_j \\ & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \quad j = 1, \dots, n \end{aligned}$$

$$\begin{aligned} \max \quad & \mathbf{c}^T \mathbf{x} \\ & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq 0 \end{aligned}$$

$$\mathbf{x} \in \mathbb{R}^n, \mathbf{c} \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m$$

$$\begin{aligned} \max \quad & 6x_1 + 8x_2 \\ & 5x_1 + 10x_2 \leq 60 \\ & 4x_1 + 4x_2 \leq 40 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$$\max \quad \begin{bmatrix} 6 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 10 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 60 \\ 40 \end{bmatrix}$$

$$x_1, x_2 \geq 0$$

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Duality

Resource Valuation problem: Determine the value of the raw materials in hand such that:

(i) it would be convenient selling and (ii) an outside company would be willing to buy them.

- z_i value of a unit of raw material i
- $\sum_{i=1}^m b_i z_i$ total expenses for buying (or opportunity cost, cost of having instead of selling)
- ρ_i prevailing unit market value of material i
- σ_j prevailing unit product price

Goal: for the outside company to minimize the total expenses;

(for the owing company the minimum amount of opportunity cost to accept for selling)

$$\min \sum_{i=1}^m b_i z_i \tag{1}$$

$$\sum_{i=1}^m z_i a_{ij} \geq \sigma_j, \quad j = 1 \dots n \tag{2}$$

$$z_i \geq \rho_i, \quad i = 1 \dots m \tag{3}$$

(2) otherwise not selling and (3) otherwise selling to someone else

Let

$$y_i = z_i - \rho_i$$

markup that the company would make by selling the raw material instead of producing.

$$\begin{aligned} \min \quad & \sum_{i=1}^m y_i b_i + \cancel{\sum_i \rho_i b_i} \\ & \sum_{i=1}^m y_i a_{ij} \geq c_j, \quad j = 1 \dots n \\ & y_i \geq 0, \quad i = 1 \dots m \end{aligned}$$

Dual Problem

$$\begin{aligned} \max \quad & \sum_{j=1}^n c_j x_j \\ & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \quad j = 1, \dots, n \end{aligned}$$

Primal Problem

Notions of Computer Science

Algorithm: a finite, well-defined sequence of operations to perform a calculation

Algorithm: LargestNumber

Input: A non-empty list of numbers L

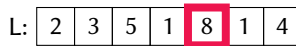
Output: The largest number in the list L

largest $\leftarrow$ L[0]

foreach each item in the list L **do**

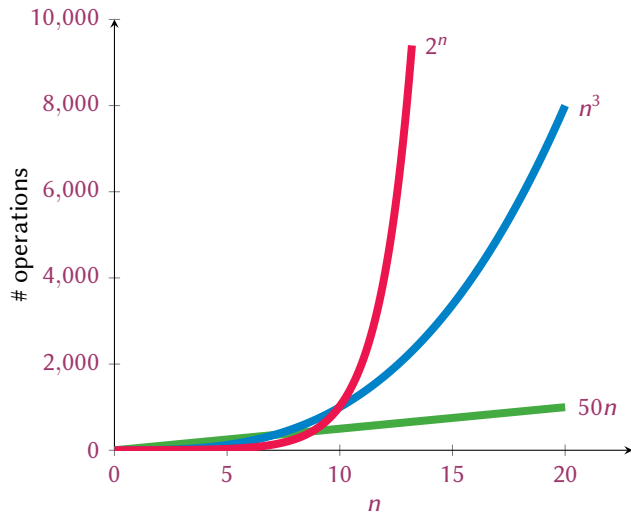
if the item > largest **then**
 largest $\leftarrow$ the item

return largest



Running time: proportional to number of operations, eg $O(n)$

Growth Functions



NP-hard problems: bad if we have to solve them, good for cryptography

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14. Sensitivity Analysis

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The Diet Problem (Blending Problems)

- Select a set of foods that will satisfy a set of daily nutritional requirements at minimum cost.
- Motivated in the 1930s and 1940s by US army.
- Formulated as a **linear programming problem** by George Stigler
- (programming intended as planning not computer code)

min cost/weight

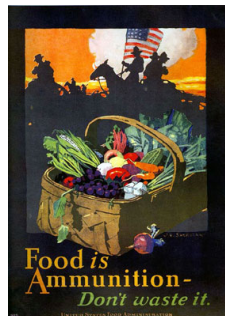
subject to nutrition requirements:

eat enough but not too much of Vitamin A

eat enough but not too much of Sodium

eat enough but not too much of Calories

...



The Diet Problem

Suppose there are:

- 3 foods available: corn, milk, and bread, and
- there are restrictions on the number of calories (between 2000 and 2250) and the amount of Vitamin A (between 5,000 and 50,000)

Food	Cost per serving	Vitamin A	Calories
Corn	\$0.18	107	72
2% Milk	\$0.23	500	121
Wheat Bread	\$0.05	0	65

The Mathematical Model

Parameters (given data)

F := set of foods

N := set of nutrients

a_{ij} := amount of nutrient i in food j , $\forall i \in N, \forall j \in F$

c_j := cost per serving of food j , $\forall j \in F$

$F_{min,j}$:= minimum number of required servings of food j , $\forall j \in F$

$F_{max,j}$:= maximum allowable number of servings of food j , $\forall j \in F$

$N_{min,i}$:= minimum required level of nutrient i , $\forall i \in N$

$N_{max,i}$:= maximum allowable level of nutrient i , $\forall i \in N$

Decision Variables

x_j := number of servings of food j to purchase/consume, $\forall j \in F$

The Mathematical Model

Objective Function: Minimize the total cost of the food

$$\text{Minimize } \sum_{j \in F} c_j x_j$$

Constraint Set 1: For each nutrient $i \in N$, at least meet the minimum required level

$$\sum_{j \in F} a_{ij} x_j \geq N_{\min, i}, \quad \forall i \in N$$

Constraint Set 2: For each nutrient $i \in N$, do not exceed the maximum allowable level.

$$\sum_{j \in F} a_{ij} x_j \leq N_{\max, i}, \quad \forall i \in N$$

Constraint Set 3: For each food $j \in F$, select at least the minimum required number of servings

$$x_j \geq F_{\min, j}, \quad \forall j \in F$$

Constraint Set 4: For each food $j \in F$, do not exceed the maximum allowable number of servings.

$$x_j \leq F_{\max, j}, \quad \forall j \in F$$

The Mathematical Model

system of equalities and inequalities

$$\min \sum_{j \in F} c_j x_j$$

$$\sum_{j \in F} a_{ij} x_j \geq N_{\min, i}, \quad \forall i \in N$$

$$\sum_{j \in F} a_{ij} x_j \leq N_{\max, i}, \quad \forall i \in N$$

$$x_j \geq F_{\min, j}, \quad \forall j \in F$$

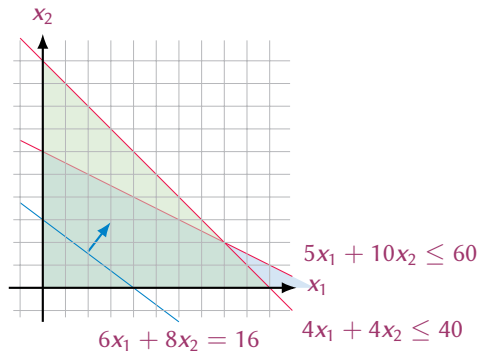
$$x_j \leq F_{\max, j}, \quad \forall j \in F$$

Mathematical Model

Machines/Materials A and B
Products 1 and 2

$$\begin{aligned} \max \quad & 6x_1 + 8x_2 \\ & 5x_1 + 10x_2 \leq 60 \\ & 4x_1 + 4x_2 \leq 40 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{aligned}$$

Graphical Representation:



In Matrix Form

$$\begin{array}{ll}\max & c_1x_1 + c_2x_2 + c_3x_3 + \dots + c_nx_n = z \\ \text{s.t.} & a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n \leq b_1 \\ & a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n \leq b_2 \\ & \dots \\ & a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n \leq b_m \\ & x_1, x_2, \dots, x_n \geq 0\end{array}$$

$$\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\begin{array}{ll}\max & z = \mathbf{c}^T \mathbf{x} \\ & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq 0\end{array}$$

Linear Programming

Abstract mathematical model:

Parameters, Decision Variables, Objective, Constraints
(+ Domains & Quantifiers)

The Syntax of a Linear Programming Problem

objective func.
constraints

$$\begin{array}{ll} \max / \min & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & A\mathbf{x} \begin{array}{l} \geq \\ \leq \\ = \end{array} \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{array}$$

$$\begin{array}{l} \mathbf{c} \in \mathbb{R}^n \\ A \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m \\ \mathbf{x} \in \mathbb{R}^n, \mathbf{0} \in \mathbb{R}^n \end{array}$$

Essential features: continuity, linearity (proportionality and additivity), certainty of parameters

- Any vector $\mathbf{x} \in \mathbb{R}^n$ satisfying all constraints is a **feasible solution**. $\mathbf{x} \in F$, F feasibility region.
- Each $\mathbf{x}^* \in \mathbb{R}^n$ that gives the best possible value for $\mathbf{c}^T \mathbf{x}$ among all feasible $\mathbf{x}$ is an **optimal solution** or **optimum**: $\mathbf{x}^*$ is optimum iff $\nexists \mathbf{x} \in F \mid \mathbf{c}^T \mathbf{x} < \mathbf{c}^T \mathbf{x}^*$
- The value $\mathbf{c}^T \mathbf{x}^*$ is the **optimum value**

Diet Problem — History

- The linear programming model consisted of 9 equations in 77 variables
- In 1944, Stigler guessed a near-optimal solution using a heuristic method
- In 1947, the National Bureau of Standards used the newly developed simplex method to solve Stigler's model.
It took 9 clerks using hand-operated desk calculators 120 man days to solve for the optimal solution
- The original instance:
https://developers.google.cn/optimization/lp/stigler_diet

AMPL Model

```
# diet.mod
set NUTR;
set FOOD;

param cost {FOOD} > 0;
param f_min {FOOD} >= 0;
param f_max { j in FOOD } >= f_min[j];
param n_min { NUTR } >= 0;
param n_max { i in NUTR } >= n_min[i];
param amt {NUTR,FOOD} >= 0;

var Buy { j in FOOD } >= f_min[j], <= f_max[j]

minimize total_cost: sum { j in FOOD } cost [j] * Buy[j];
subject to diet { i in NUTR }:
    n_min[i] <= sum { j in FOOD } amt[i,j] * Buy[j] <= n_max[i];
```


AMPL Model

```
# diet.dat
data;

set NUTR := A B1 B2 C ;
set FOOD := BEEF CHK FISH HAM MCH MIL SPG TUR;

param: cost f_min f_max :=
  BEEF 3.19 0 100
  CHK 2.59 0 100
  FISH 2.29 0 100
  HAM 2.89 0 100
  MCH 1.89 0 100
  MIL 1.99 0 100
  SPG 1.99 0 100
  TUR 2.49 0 100 ;

param: n_min n_max :=
  A 700 10000
  C 700 10000
  B1 700 10000
  B2 700 10000 ;

# %
```

```
param amt (tr):
      A    C    B1    B2 :=
  BEEF 60  20  10  15
  CHK  8   0  20  20
  FISH 8   10  15  10
  HAM  40  40  35  10
  MCH  15  35  15  15
  MIL  70  30  15  15
  SPG  25  50  25  15
  TUR  60  20  15  10 ;
```

Python Script

Model

```
# Model diet.py
m = Model("diet")

# Create decision variables for the foods to buy
buy = {}
for f in foods:
    buy[f] = m.addVar(obj=cost[f], name=f)

# Nutrition constraints
for c in categories:
    m.addConstr(
        quicksum(nutritionValues[f,c] * buy[f] for f in foods) <= maxNutrition[c], name=c+'max')
    m.addConstr(
        quicksum(nutritionValues[f,c] * buy[f] for f in foods) >= minNutrition[c], name=c+'min')

# Solve
m.optimize()
```

Python Script

Data

```
import gurobipy as gp

categories, minNutrition, maxNutrition = gp.multidict({
    'calories': [1800, 2200],
    'protein': [91, gp.GRB.INFINITY],
    'fat': [0, 65],
    'sodium': [0, 1779] })

foods, cost = gp.multidict({
    'hamburger': 2.49,
    'chicken': 2.89,
    'hot dog': 1.50,
    'fries': 1.89,
    'macaroni': 2.09,
    'pizza': 1.99,
    'salad': 2.49,
    'milk': 0.89,
    'ice cream': 1.59 })
```

Nutrition values for the foods

```
nutritionValues = {
    ('hamburger', 'calories'): 410,
    ('hamburger', 'protein'): 24,
    ('hamburger', 'fat'): 26,
    ('hamburger', 'sodium'): 730,
    ('chicken', 'calories'): 420,
    ('chicken', 'protein'): 32,
    ('chicken', 'fat'): 10,
    ('chicken', 'sodium'): 1190,
    ('hot dog', 'calories'): 560,
    ('hot dog', 'protein'): 20,
    ('hot dog', 'fat'): 32,
    ('hot dog', 'sodium'): 1800,
    ('fries', 'calories'): 380,
    ('fries', 'protein'): 4,
    ('fries', 'fat'): 19,
    ('fries', 'sodium'): 270,
    ('macaroni', 'calories'): 320,
```

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History of Linear Programming (LP)

System of linear equations

↪ It is impossible to find out who knew what, when, first.

Just two “references”:

- Egyptians and Babylonians considered about 2000 B.C. the solution of special linear equations. But, of course, they described examples and did not describe the methods in "today's style".
- What we call “**Gaussian elimination**” today has been explicitly described in Chinese “Nine Books of Arithmetics” which is a compendium written in the period 2000 B.C. to A.D. 9, but the methods were probably known long before that.
- **Gauss**, by the way, never described “Gaussian elimination”. He just used it and stated that the linear equations he used can be solved “per eliminationem vulgarem”

History of Linear Programming (LP)

George B. Dantzig, (2002) Linear Programming. Operations Research 50(1):42-47.

<https://doi.org/10.1287/opre.50.1.42.17798>

- Origins of LP date back to **Newton**, **Leibnitz**, **Lagrange**, etc.
- In 1827, **Fourier** described a variable elimination method for **systems of linear inequalities**, today often called Fourier-Motzkin elimination (Motzkin, 1937). It can be turned into an LP solver but inefficient.
- In 1932, **Leontief** (1905-1999) Input-Output model to represent interdependencies between branches of a national economy (1976 Nobel prize)
- In 1939, **Kantorovich** (1912-1986): Foundations of linear programming (Nobel prize in economics with **Koopmans** on LP, 1975) on Optimal use of scarce resources: foundation and economic interpretation of LP
- The math subfield of **Linear Programming** was created by **George Dantzig**, **John von Neumann** (Princeton), and **Leonid Kantorovich** in the 1940s.

History of LP (cntd)

- In 1947, **Dantzig** (1914-2005) invented the **(primal) simplex algorithm** working for the US Air Force at the Pentagon. (program=plan)
- In 1954, **Lemke**: dual simplex algorithm,
- In 1954, **Dantzig** and **Orchard Hays**: revised simplex algorithm
- In 1970, **Victor Klee** and **George Minty** created an example that showed that the classical simplex algorithm has exponential worst-case behavior.
- In 1979, **L. Khachain** found a new **efficient** algorithm for linear programming. It was terribly slow. (Ellipsoid method)
- In 1984, **Karmarkar** discovered yet another new **efficient** algorithm for linear programming. It proved to be a strong competitor for the simplex method. (Interior point method)

History of Optimization

- In 1951, Nonlinear Programming began with the Karush-Kuhn-Tucker Conditions
- In 1952, Commercial Applications and Software began
- In 1950s, Network Flow Theory began with the work of Ford and Fulkerson.
- In 1955, Stochastic Programming began
- In 1958, Integer Programming began by R.E. Gomory.
- In 1962, Complementary Pivot Theory

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Fourier Motzkin elimination method

Has $\mathbf{Ax} \leq \mathbf{b}$ a solution? (Assumption: $\mathbf{A} \in \mathbb{Q}^{m \times n}$, $\mathbf{b} \in \mathbb{Q}^n$)

Idea:

1. transform the system into another by eliminating some variables such that the two systems have the same solutions over the remaining variables.
2. reduce to a system of constant inequalities that can be easily decided

Let x_r be the variable to eliminate

Let $M = \{1 \dots m\}$ indices of the constraints

For a variable j let's partition the rows of the matrix in

$$N = \{i \in M \mid a_{ij} < 0\}$$

$$Z = \{i \in M \mid a_{ij} = 0\}$$

$$P = \{i \in M \mid a_{ij} > 0\}$$

$$\left\{ \begin{array}{ll} x_r \geq b'_{ir} - \sum_{k=1}^{r-1} a'_{ik} x_k, & a_{ir} < 0 \\ x_r \leq b'_{ir} - \sum_{k=1}^{r-1} a'_{ik} x_k, & a_{ir} > 0 \\ \text{all other constraints} & i \in Z \end{array} \right. \quad \left\{ \begin{array}{ll} x_r \geq A_i(x_1, \dots, x_{r-1}), & i \in N \\ x_r \leq B_i(x_1, \dots, x_{r-1}), & i \in P \\ \text{all other constraints} & i \in Z \end{array} \right.$$

Hence the original system is equivalent to

$$\left\{ \begin{array}{l} \max\{A_i(x_1, \dots, x_{r-1}), i \in N\} \leq x_r \leq \min\{B_i(x_1, \dots, x_{r-1}), i \in P\} \\ \text{all other constraints} \quad i \in Z \end{array} \right.$$

which is equivalent to

$$\left\{ \begin{array}{ll} A_i(x_1, \dots, x_{r-1}) \leq B_j(x_1, \dots, x_{r-1}) & i \in N, j \in P \\ \text{all other constraints} & i \in Z \end{array} \right.$$

we eliminated x_r but:

$$\left\{ \begin{array}{l} |N| \cdot |P| \text{ inequalities} \\ |Z| \text{ inequalities} \end{array} \right.$$

after d iterations if $|P| = |N| = m/2$ exponential growth: $(1/4^d)(m/2)^{2^d}$

Example

$$\begin{array}{rcl} -7x_1 & + & 6x_2 \leq 25 \\ x_1 & - & 5x_2 \leq 1 \\ x_1 & & \leq 7 \\ -x_1 & + & 2x_2 \leq 12 \\ -x_1 & - & 3x_2 \leq 1 \\ 2x_1 & - & x_2 \leq 10 \end{array}$$

x_2 variable to eliminate

$$N = \{2, 5, 6\}, Z = \{3\}, P = \{1, 4\}$$

$$|Z \cup (N \times P)| = 7 \text{ constraints}$$

By adding one variable and one inequality, Fourier-Motzkin elimination can be turned into an LP solver.

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Definitions

- $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$ closed interval
 $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$ open interval
- column vectors and matrices
scalar product: $\mathbf{y}^T \mathbf{x} = \sum_{i=1}^n y_i x_i$
- $A\mathbf{x}$ column vector combination of the columns of A ;
 $\mathbf{u}^T A$ row vector combination of the rows of A
- **linear combination**

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$$
$$\boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_k]^T \in \mathbb{R}^k$$

$$\mathbf{x} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_k \mathbf{v}_k = \sum_{i=1}^k \lambda_i \mathbf{v}_i$$

moreover:

$$\boldsymbol{\lambda} \geq \mathbf{0}$$

$$\boldsymbol{\lambda}^T \mathbf{1} = 1$$

$$\boldsymbol{\lambda} \geq \mathbf{0} \quad \text{and} \quad \boldsymbol{\lambda}^T \mathbf{1} = 1$$

conic combination

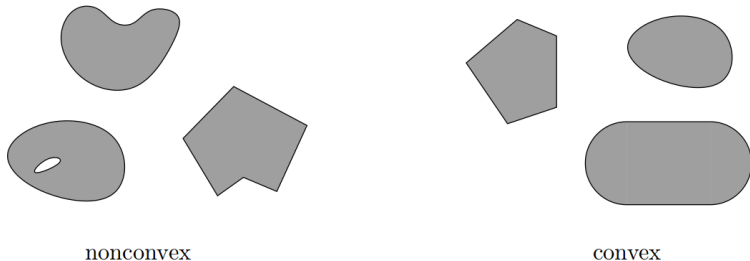
affine combination

convex combination

$$\left(\sum_{i=1}^k \lambda_i = 1 \right)$$

Definitions

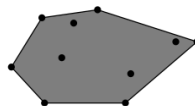
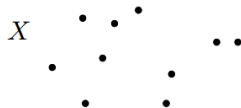
- set S is **linear (affine) independent** if no element of it can be expressed as linear combination of the others
Eg: $S \subseteq \mathbb{R}^n \implies \max n \text{ lin. indep. (max } n+1 \text{ aff. indep.)}$
- **convex set**: if $\mathbf{x}, \mathbf{y} \in S$ and $0 \leq \lambda \leq 1$ then $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in S$



- $f : X \rightarrow \mathbb{R}$ is a **convex function** if its epigraph $\{(x, y) \in \mathbb{R}^2 : y \geq f(x)\}$ is a convex set or if $\forall \mathbf{x}, \mathbf{y} \in X, \lambda \in [0, 1]$ it holds that $f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y})$

Definitions

- For a set of points $S \subseteq \mathbb{R}^n$
 - $\text{lin}(S)$ linear hull (span)
 - $\text{cone}(S)$ conic hull
 - $\text{aff}(S)$ affine hull
 - $\text{conv}(S)$ convex hull



the convex hull of X

$$\text{conv}(X) = \{ \lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2 + \dots + \lambda_n \mathbf{x}_n \mid \mathbf{x}_i \in X, \lambda_1, \dots, \lambda_n \geq 0 \text{ and } \sum_i \lambda_i = 1 \}$$

Definitions

- **rank** of a matrix for columns (= for rows)
if (m, n) -matrix has rank = $\min\{m, n\}$ then the matrix is full rank
if (n, n) -matrix is full rank then it is regular and admits an inverse

- $G \subseteq \mathbb{R}^n$ is an **hyperplane** if $\exists \mathbf{a} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $\alpha \in \mathbb{R}$:

$$G = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}^T \mathbf{x} = \alpha\}$$

- $H \subseteq \mathbb{R}^n$ is an **halfspace** if $\exists \mathbf{a} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $\alpha \in \mathbb{R}$:

$$H = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}^T \mathbf{x} \leq \alpha\}$$

($\mathbf{a}^T \mathbf{x} = \alpha$ is a supporting hyperplane of H)

Definitions

- a set $P \subset \mathbb{R}^n$ is a **polyhedron** if $\exists m \in \mathbb{Z}^+, A \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m$:

$$P = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} \leq \mathbf{b}\} = \bigcap_{i=1}^m \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}_i^T \cdot \mathbf{x} \leq b_i\}$$

i.e., a polyhedron $P \neq \mathbb{R}^n$ is determined by finitely many halfspaces

- a polyhedron P is a **polytope** if it is bounded: $\exists B \in \mathbb{R}, B > 0$:

$$P \subseteq \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| \leq B\}$$

($\|\mathbf{x}\| = \sqrt{\sum_{i=1}^n x_i^2}$ is the Euclidean norm of the vector $\mathbf{x} \in \mathbb{R}$)

- a point $\mathbf{x}$ of a polyhedron P is said to be an **extreme point** or a **vertex** of P if it cannot be expressed as a strict convex combination of other two points of the polyhedron, i.e., if there exist no $\mathbf{y}, \mathbf{z} \in P, \mathbf{y} \neq \mathbf{z}$ and $\lambda \in (0, 1)$ such that $\mathbf{x} = \lambda \mathbf{y} + (1 - \lambda) \mathbf{z}$
- every point of a **polytope** can be obtained as the convex combination of its vertices.
(Minkowski-Weyl Theorem)

Definitions

- If A and b are made of rational numbers, $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$ is a **rational polyhedron**
- General optimization problem: $\max\{\varphi(x) \mid x \in F\}$, F is feasible region for x
- Note: if F is open, eg, $x < 5$ then: $\sup\{x \mid x < 5\}$
supremum: least element of $\mathbb{R}$ greater or equal than any element in F
- $\arg \min\{f(i) \mid i \in I\}$ argument $i^* \in I$ such that $f(i^*) = \min\{f(i) \mid i \in I\}$

Definitions

- The inequality denoted by $(\mathbf{a}, \alpha)$ is called a **valid inequality for P** if $\mathbf{a}\mathbf{x} \leq \alpha, \forall \mathbf{x} \in P$.
Note that $(\mathbf{a}, \alpha)$ is a valid inequality if and only if P lies in the half-space $\{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}\mathbf{x} \leq \alpha\}$.
- A **face** of P is $F = \{\mathbf{x} \in P \mid \mathbf{a}\mathbf{x} = \alpha\}$ where $(\mathbf{a}, \alpha)$ is a valid inequality for P . Hence, it is the intersection of P with the hyperplane of a valid inequality. It is said to be **proper** if $F \neq \emptyset$ and $F \neq P$.
- If $F \neq \emptyset$ we say that the corresponding hyperplane **supports P** .
If $\mathbf{c}$ is a non zero vector for which $\delta = \max\{\mathbf{c}^T \mathbf{x} \mid \mathbf{x} \in P\}$ is finite, then the set $\{\mathbf{x} \mid \mathbf{c}^T \mathbf{x} = \delta\}$ is called **supporting hyperplane**.
- A point $\mathbf{x}$ for which $\{\mathbf{x}\}$ is a face is called a **vertex** of P and also a **basic solution** of $\mathbf{A}\mathbf{x} \leq \mathbf{b}$ (0 dim face)
- A **facet** is a maximal face distinct from P
 $\mathbf{c}\mathbf{x} \leq d$ is facet defining if $\mathbf{c}\mathbf{x} = d$ is a supporting hyperplane of P of $n - 1$ dim

Linear Programming Problem

Input: a matrix $A \in \mathbb{R}^{m \times n}$ and column vectors $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{c} \in \mathbb{R}^n$

Task:

1. decide that $\{\mathbf{x} \in \mathbb{R}^n; A\mathbf{x} \leq \mathbf{b}\}$ is empty (prob. infeasible), or
2. find a column vector $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} \leq \mathbf{b}$ and $\mathbf{c}^T \mathbf{x}$ is max, or
3. decide that for all $\alpha \in \mathbb{R}$ there is an $\mathbf{x} \in \mathbb{R}^n$ with $A\mathbf{x} \leq \mathbf{b}$ and $\mathbf{c}^T \mathbf{x} > \alpha$ (prob. unbounded)

1. $F = \emptyset$
2. $F \neq \emptyset$ and $\exists$ solution
 1. one solution
 2. infinite solutions
3. $F \neq \emptyset$ and $\nexists$ solution

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Fundamental Theorem of LP

Theorem (Fundamental Theorem of Linear Programming)

Given:

$$\min\{\mathbf{c}^T \mathbf{x} \mid \mathbf{x} \in P\} \text{ where } P = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} \leq \mathbf{b}\}$$

If P is a bounded polyhedron and not empty and $\mathbf{x}^*$ is an optimal solution to the problem, then:

- $\mathbf{x}^*$ is an extreme point (vertex) of P , or
- $\mathbf{x}^*$ lies on a face $F \subset P$ of optimal solutions



Proof idea:

- assume $\mathbf{x}^*$ not on the boundary of P then $\exists$ a ball around it still in P . Show that a point in the ball has better cost
- if $\mathbf{x}^*$ is not unique then any convex combination of other optimal points are also optimal.

Implications:

- the optimal solution is at the intersection of supporting hyperplanes.
- hence finitely many possibilities
- solution method: write all inequalities as equalities and solve all $\binom{m}{n}$ systems of linear equalities (n # variables, m # equality constraints)
- for each point we then need to check if feasible and if best in cost.
- each system is solved by Gaussian elimination
- Stirling approximation:

$$\binom{2m}{m} \approx \frac{4^m}{\sqrt{\pi m}} \text{ as } m \rightarrow \infty$$

Simplex Method

1. find a solution that is at the intersection of some n hyperplanes
2. try systematically to produce the other points by exchanging one hyperplane with another
3. check optimality, proof provided by duality theory

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Gaussian Elimination

1. Forward elimination

reduces the system to row echelon form by elementary row operations

- multiply a row by a non-zero constant
- interchange two rows
- add a multiple of one row to another

(or LU decomposition)

2. Back substitution (or reduced row echelon form - RREF)

Example

$$\begin{aligned} 2x + y - z &= 8 & (R1) \\ -3x - y + 2z &= -11 & (R2) \\ -2x + y + 2z &= -3 & (R3) \end{aligned}$$

R1	2	1	-1	8	
R2	-3	-1	2	-11	
R3	-2	1	2	-3	

$$\begin{aligned} 2x + y - z &= 8 & (R1) \\ + \frac{1}{2}y + \frac{1}{2}z &= 1 & (R2) \\ + 2y + 1z &= 5 & (R3) \end{aligned}$$

R1'	= 1/2 R1				
R2'	= R2 + 3/2 x R1				
R3'	= R3 + R1				

$$\begin{aligned} 2x + y - z &= 8 & (R1) \\ + \frac{1}{2}y + \frac{1}{2}z &= 1 & (R2) \\ - z &= 1 & (R3) \end{aligned}$$

R1'	= R1				
R2'	= 2 R2				
R3'	= R3 - 4 x R2				

$$\begin{aligned} 1x + 1/2y &= 7/2 & (R1) \\ 1y &= 3 & (R2) \\ z &= -1 & (R3) \end{aligned}$$

R1'	= R1 - 1/2 x R3				
R2'	= R2 + R3				
R3'	= -R3				

$$\begin{aligned} x &= 2 & (R1) \\ y &= 3 & (R2) \\ z &= -1 & (R3) \end{aligned}$$

R1'	= R1 - 1/2 x R2				
R2'	= R2				
R3'	= R3				

=> x=2

=> y=3

=> z=-1

LU Factorization

$$\begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -11 \\ -3 \end{bmatrix}$$

$$A\mathbf{x} = \mathbf{b}$$

$$\mathbf{x} = A^{-1}\mathbf{b}$$

$$\begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$A = PLU$$

$$\mathbf{x} = A^{-1}\mathbf{b} = U^{-1}L^{-1}P^T\mathbf{b}$$

$$\mathbf{z}_1 = P^T\mathbf{b}, \quad \mathbf{z}_2 = L^{-1}\mathbf{z}_1, \quad \mathbf{x} = U^{-1}\mathbf{z}_2$$

In Python

```
In [1]: import scipy as sc
```

```
In [2]: A = sc.array([[2,1,-1],[-3,-1,2],[-2,1,2]])
```

```
In [3]: from scipy import linalg as sl
```

```
In [4]: P,L,U = sl.lu(A)
```

```
In [5]: print(P,L,U)
```

```
[[0. 0. 1.]  
 [1. 0. 0.]  
 [0. 1. 0.]]  
[[ 1.          0.          0.          ]  
 [ 0.66666667  1.          0.          ]  
 [-0.66666667  0.2         1.          ]]  
[[-3.         -1.         2.          ]  
 [ 0.          1.66666667  0.66666667]  
 [ 0.          0.          0.2         ]]
```

Polynomial time $O(n^2m)$ but needs to guarantee that all the numbers during the run can be represented by polynomially bounded bits

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A Numerical Example

$$\begin{aligned} \max \quad & \sum_{j=1}^n c_j x_j \\ & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \quad j = 1, \dots, n \end{aligned}$$

$$\begin{aligned} \max \quad & \mathbf{c}^T \mathbf{x} \\ & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

$$\mathbf{x} \in \mathbb{R}^n, \mathbf{c} \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m$$

$$\begin{aligned} \max \quad & 6x_1 + 8x_2 \\ & 5x_1 + 10x_2 \leq 60 \\ & 4x_1 + 4x_2 \leq 40 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$$\max \quad \begin{bmatrix} 6 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 10 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 60 \\ 40 \end{bmatrix}$$

$$x_1, x_2 \geq 0$$

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Standard Form

Every LP problem can be converted in the **standard form**:

$$\begin{aligned} \max \quad & \mathbf{c}^T \mathbf{x} \\ & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \in \mathbb{R}^n \end{aligned}$$

$$\mathbf{c} \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m$$

- if equations, then put two constraints, $\mathbf{ax} \leq b$ and $\mathbf{ax} \geq b$
- if $\mathbf{ax} \geq b$ then $-\mathbf{ax} \leq -b$
- if $\min \mathbf{c}^T \mathbf{x}$ then $-\max(-\mathbf{c}^T \mathbf{x})$

and then be put in **equational standard form**:

$$\begin{aligned} \max \quad & \mathbf{c}^T \mathbf{x} \\ & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

$$\mathbf{x} \in \mathbb{R}^n, \mathbf{c} \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m$$

1. “=” constraints
2. $\mathbf{x} \geq 0$ nonnegativity constraints
3. ($\mathbf{b} \geq 0$)
4. max

Transformation to Std Form

Every LP problem can be transformed in eq. std. form

1. introduce slack variables (or surplus)

$$5x_1 + 10x_2 + x_3 = 60$$

$$4x_1 + 4x_2 + x_4 = 40$$

2. if $x_1 \begin{matrix} \geq \\ \leq \end{matrix} 0$ then
$$\begin{aligned} x_1 &= x'_1 - x''_1 \\ x'_1 &\geq 0 \\ x''_1 &\geq 0 \end{aligned}$$

3. ($b \geq 0$)

4. $\min c^T x \equiv -\max(-c^T x)$

LP in $m \times n$ converted into LP with at most $(m + 2n)$ variables and m equations (n # original variables, m # constraints)

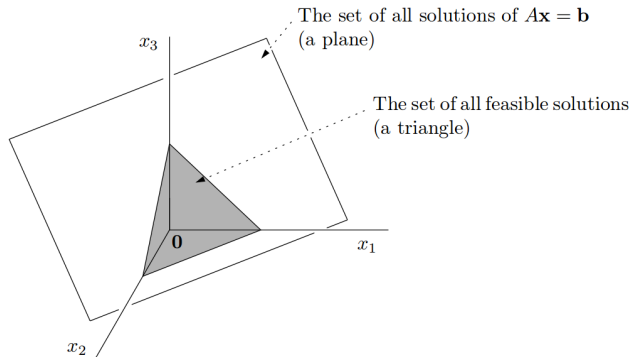
Geometry of LP in Eq. Std. Form

$$\max\{\mathbf{c}^T \mathbf{x} \mid A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$$

In $\mathbb{R}^3$:

From linear algebra:

- the set of solutions of $A\mathbf{x} = \mathbf{b}$ is an affine space (hyperplane not passing through the origin).
- $\mathbf{x} \geq \mathbf{0}$ nonnegative orthant (octant in $\mathbb{R}^3$)



- $A\mathbf{x} = \mathbf{b}$ is a system of equations that we can solve by Gaussian elimination
- Elementary row operations of $[A \mid \mathbf{b}]$ do not affect set of feasible solutions
 - multiplying all entries in some row of $[A \mid \mathbf{b}]$ by a nonzero real number λ
 - replacing the i th row of $[A \mid \mathbf{b}]$ by the sum of the i th row and j th row for some $i \neq j$
- Let n' be the number of vars in eq. std. form.

we assume $n' \geq m$ and $\text{rank}([A \mid \mathbf{b}]) = \text{rank}(A) = m$

ie, rows of A are linearly independent
 otherwise, remove linear dependent rows

If $A \in \mathbb{R}^{m \times n'}$ has rank m , then $A\mathbf{x} = \mathbf{b}$ is an $(n' - m)$ -dimensional affine plane.

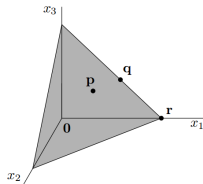
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Basic Feasible Solutions

Basic feasible solutions are the vertices of the feasible region:



More formally:

Let $B = \{1 \dots m\}$, $N = \{m+1 \dots n+m = n'\}$ be subsets partitioning the columns of A : A_B be made of columns of A indexed by B :

Definition

$\mathbf{x} \in \mathbb{R}^n$ is a **basic feasible solution** of the linear program $\max\{\mathbf{c}^T \mathbf{x} \mid A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$ for an index set B if:

- $x_j = 0 \ \forall j \notin B$
- the square matrix A_B is nonsingular, ie, all columns indexed by B are lin. indep.
- $\mathbf{x}_B = A_B^{-1} \mathbf{b}$ is nonnegative, ie, $\mathbf{x}_B \geq \mathbf{0}$ (feasibility)

We call x_j for $j \in B$ **basic variables** and remaining variables **nonbasic variables**.

Theorem

A **basic feasible solution** is uniquely determined by the set B .

Proof:

$$A\mathbf{x} = A_B\mathbf{x}_B + A_N\mathbf{x}_N = \mathbf{b}$$

$$\mathbf{x}_B + A_B^{-1}A_N\mathbf{x}_N = A_B^{-1}\mathbf{b}$$

$$\mathbf{x}_B = A_B^{-1}\mathbf{b}$$

A_B is nonsingular hence one solution

Note: we call B a **(feasible) basis**

Definition

A **basic feasible solution** of a linear program with n variables is a feasible solution for which some n linearly independent constraints hold with equality.

Extreme points and basic feasible solutions are geometric and algebraic manifestations of the same concept:

Theorem

Let P be a (convex) polyhedron made of all feasible solutions of an LP in std. form. For a point $v \in P$ the following are equivalent:

- (i) v is an extreme point (vertex) of P*
- (ii) v is a basic feasible solution of LP*

Proof: see text book [MG] sec. 4.4.

Theorem

Let $LP = \max\{\mathbf{c}^T \mathbf{x} \mid A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$ be feasible and bounded, then there is an optimal solution that is a basic feasible solution.

Proof. consequence of previous theorem and fundamental theorem of linear programming

A similar theorem is valid for arbitrary linear programs (not in eq. form)

However, an optimal solution does not need to be basic:

$$\max x_1 + x_2 \text{ subject to } x_1 + x_2 \leq 1$$

(bounded case with infinite solutions)

- Idea for solution method:
- examine all basic solutions.
- There are finitely many: $\binom{m+n}{m}$.
- However, if $n = m$ then $\binom{2m}{m} \approx 4^m$.

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Simplex Method

$$\max \quad z = [6 \ 8] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 10 & 1 & 0 \\ 4 & 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 60 \\ 40 \end{bmatrix}$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Canonical eq. std. form: one decision variable is isolated in each constraint with coefficient 1 and does not appear in the other constraints nor in the obj. func. and b terms are positive

It gives immediately a basic feasible solution:

$$x_1 = 0, x_2 = 0, x_3 = 60, x_4 = 40$$

Is it optimal? Look at signs in $z \rightsquigarrow$ if positive then an increase would improve.

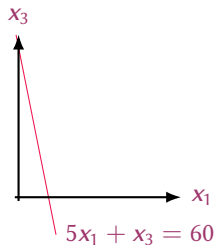
Let's try to increase a promising variable, ie, x_1 , one with positive coefficient in z

$$5x_1 + x_3 = 60$$

$$x_1 = \frac{60}{5} - \frac{x_3}{5}$$

$$x_3 = 60 - 5x_1 \geq 0$$

If $x_1 > 12$ then $x_3 < 0$

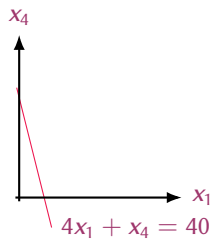


$$4x_1 + x_4 = 40$$

$$x_1 = \frac{40}{4} - \frac{x_4}{4}$$

$$x_4 = 40 - 4x_1 \geq 0$$

If $x_1 > 10$ then $x_4 < 0$



we can take the minimum of the two $\rightsquigarrow x_1$ increased to 10

x_1 **enters** the basis and x_4 **leaves** it.

Dictionary Form

Hence, we increase x_1 to 10 and consequently set $x_4 = 0$
That is, x_1 **enters** the basis and x_4 **leaves** it.

$$\begin{array}{llll} \max & 6x_1 & + & 8x_2 \\ & 5x_1 & + & 10x_2 + x_3 & = & 60 \\ & 4x_1 & + & 4x_2 & + & x_4 = 40 \\ & & & x_1, x_2 & \geq & 0 \end{array}$$

$$\begin{array}{rcl} x_3 & = & 60 - 5x_1 - 10x_2 \\ x_4 & = & 40 - 4x_1 - 4x_2 \\ \hline z & = & + 6x_1 + 8x_2 \end{array}$$

$$\begin{array}{rcl} x_3 & = & 10 - 5x_2 + 5/4x_4 \\ x_1 & = & 10 - x_2 - 1/4x_4 \\ \hline z & = & 60 + 2x_2 - 6/4x_4 \end{array}$$

Simplex Tableau

First simplex tableau:

	x_1	x_2	x_3	x_4	$-z$	b
x_3	5	10	1	0	0	60
x_4	4	4	0	1	0	40
	6	8	0	0	1	0

we want to reach this new tableau

	x_1	x_2	x_3	x_4	$-z$	b
x_3	0	?	1	?	0	?
x_1	1	?	0	?	0	?
	0	?	0	?	1	?

Pivot operation:

1. Choose pivot:

column: one s with positive coefficient in obj. func.

row: ratio between coefficient b and pivot column: choose the one with smallest ratio:

$$\theta = \min_i \left\{ \frac{b_i}{a_{is}} : a_{is} > 0 \right\}, \quad \theta \text{ increase value of entering var.}$$

2. elementary row operations to update the tableau

- x_4 leaves the basis, x_1 enters the basis
 - Divide pivot row by pivot
 - Send to zero the coefficient in the pivot column of the first row
 - Send to zero the coefficient of the pivot column in the third (cost) row

	x_1	x_2	x_3	x_4	$-z$	b
I' = I - 5II'	0	5	1	-5/4	0	10
II' = II/4	1	1	0	1/4	0	10
III' = III - 6II'	0	2	0	-6/4	1	-60

From the last row we read: $2x_2 - 3/2x_4 - z = -60$, that is: $z = 60 + 2x_2 - 3/2x_4$.
 Since x_2 and x_4 are nonbasic we have $z = 60$ and $x_1 = 10, x_2 = 0, x_3 = 10, x_4 = 0$.

- Done? No! Let x_2 enter the basis

	x_1	x_2	x_3	x_4	$-z$	b
I' = I/5	0	1	1/5	-1/4	0	2
II' = II - I'	1	0	-1/5	1/2	0	8
III' = III - 2I'	0	0	-2/5	-1	1	-64

Definition (Reduced costs)

We call **reduced costs** the coefficients in the objective function of the nonbasic variables, $\bar{c}_N$

Proposition (Optimality Condition)

*The basic feasible solution is **optimal** when the **reduced costs** in the corresponding simplex tableau are **nonpositive**, ie, such that:*

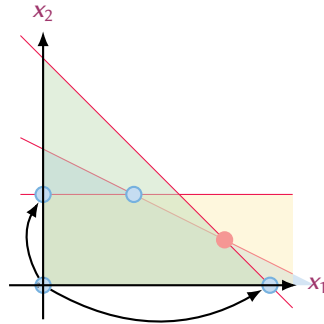
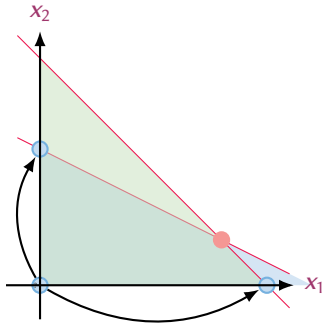
$$\bar{c}_N \leq 0$$

Proof: Let z_0 be the obj value when $\bar{c}_N \leq 0$.

For any other feasible solution $\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}_B, \tilde{\mathbf{x}}_N]$, $\tilde{\mathbf{x}}_N \geq 0$ reachable from the tableau we have:

$$z = \mathbf{c}^T \tilde{\mathbf{x}} \quad \text{and from the last line of the tableau} \quad \mathbf{c}^T \tilde{\mathbf{x}} = z = z_0 + \bar{\mathbf{c}}_B^T \tilde{\mathbf{x}}_B + \bar{\mathbf{c}}_N^T \tilde{\mathbf{x}}_N \leq z_0$$

Graphical Representation



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Tableaux and Dictionaries

$$\begin{aligned} \max \quad & \sum_{j=1}^n c_j x_j \\ & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \quad j = 1, \dots, n \end{aligned}$$

$$\begin{aligned} x_{n+i} &= b_i - \sum_{j=1}^n a_{ij} x_j, \quad i = 1, \dots, m \\ z &= \sum_{j=1}^n c_j x_j \end{aligned}$$

Tableau

$$\left[\begin{array}{c|c|c|c} I & \bar{A}_N & 0 & \bar{b} \\ \hline 0 & \bar{c}_N & 1 & -d \end{array} \right]$$

Dictionary

$$\begin{aligned} x_r &= \bar{b}_r - \sum_{s \notin B} \bar{a}_{rs} x_s, \quad r \in B \\ z &= \bar{d} + \sum_{s \notin B} \bar{c}_s x_s \end{aligned}$$

pivot operations in dictionary form:

choose col s with r.c. > 0

choose row with $\min\{-\bar{b}_i/\bar{a}_{is} \mid a_{is} < 0, i = 1, \dots, m\}$

update: express entering variable and substitute in other rows

Example

$$\begin{aligned}
 \max \quad & 6x_1 + 8x_2 \\
 & 5x_1 + 10x_2 \leq 60 \\
 & 4x_1 + 4x_2 \leq 40 \\
 & x_1, x_2 \geq 0
 \end{aligned}$$

	x_1	x_2	x_3	x_4	$-z$	b
x_3	5	10	1	0	0	60
x_4	4	4	0	1	0	40
	6	8	0	0	1	0

After 2 iterations:

	x_1	x_2	x_3	x_4	$-z$	b
x_2	0	1	$1/5$	$-1/4$	0	2
x_1	1	0	$-1/5$	$1/2$	0	8
	0	0	$-2/5$	-1	1	-64

$$x_3 = 60 - 5x_1 - 10x_2$$

$$x_4 = 40 - 4x_1 - 4x_2$$

$$z = 64 - 2x_1 - 4x_2$$

$$x_2 = 2 - 1/5x_3 + 1/4x_4$$

$$x_1 = 8 + 1/5x_3 - 1/2x_4$$

$$z = 64 - 2/5x_3 - 1x_4$$

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Simplex Algorithm so far

while there exists a nonbasic variable with **positive** reduced cost **do**

1. Choose pivot:

column: one s with positive coefficient in obj. func. later)
($\equiv$ choosing the entering variable)

row: one with the smallest ratio between coefficient b and
coefficient in the pivot column (aka **minimum ratio test**):

$$\theta = \min_i \left\{ \frac{b_i}{a_{is}} : a_{is} > 0 \right\}, \quad \theta \text{ increase value of entering var.}$$

($\equiv$ choosing the leaving variable)

2. Elementary row operations to update the tableau

Simplex: Exception Handling, Overview

Solution of an LP problem:

- a. $F \neq \emptyset$ and $\nexists$ solution
- b. $F \neq \emptyset$ and $\exists$ solution
 - i) **one solution**
 - ii) infinite solutions
- c. $F = \emptyset$

Handling exceptions in the Simplex Method

- 1. Unboundedness
- 2. More than one solution
- 3. Degeneracies
 - benign
 - cycling
- 4. Infeasible starting
Phase I + Phase II

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Unboundedness

$$\begin{aligned}
 \max \quad & 2x_1 + x_2 \\
 & x_2 \leq 5 \\
 & -x_1 + x_2 \leq 1 \\
 & x_1, x_2 \geq 0
 \end{aligned}$$

- Initial tableau

	x1	x2	x3	x4	-z	b
x3	0	1	1	0	0	5
x4	-1	1	0	1	0	1
	2	1	0	0	1	0

- x_2 entering, x_4 leaving

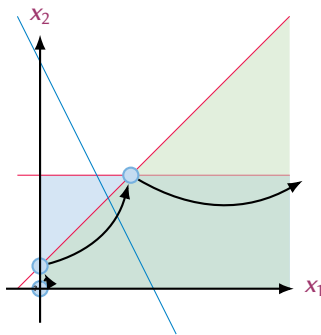
	x1	x2	x3	x4	-z	b
II' = II - I'	1	0	1	-1	0	4
I' = I	-1	1	0	1	0	1
III' = III - I'	3	0	0	-1	1	-1

$-x_1 + x_2 + x_4 = 1$, x_1 can increase without restriction, $\theta = \min\{\frac{b_i}{a_{is}} : a_{is} > 0, i = 1 \dots, n\}$

- x_1 entering, x_3 leaving

	x_1	x_2	x_3	x_4	$-z$	b
I' = I	1	0	1	-1	0	4
II' = II + I'	0	1	1	0	0	5
III' = III - 3I'	0	0	-3	2	1	-13

x_4 was already in basis but for both I and II ($x_2 + x_3 + 0x_4 = 5$), x_4 can increase arbitrarily



$$\begin{aligned}
 \max \quad & x_1 + x_2 \\
 & 5x_1 + 10x_2 \leq 60 \\
 & 4x_1 + 4x_2 \leq 40 \\
 & x_1, x_2 \geq 0
 \end{aligned}$$

- Initial tableau

	x1	x2	x3	x4	-z	b
x3	5	10	1	0	0	60
x4	4	4	0	1	0	40
	1	1	0	0	1	0

- x_2 enters, x_3 leaves

	x1	x2	x3	x4	-z	b
I' = I/10	1/2	1	1/10	0	0	6
II' = II - 4Ix4	2	0	-2/5	1	0	16
III' = III - I	1/2	0	-1/6	0	1	-6

- x_1 enters, x_4 leaves

	x_1	x_2	x_3	x_4	$-z$	b
I' = I - II' / 2	0	1	1/5	-1/4	0	2
II' = II / 2	1	0	-1/5	1/2	0	8
III' = III - II' / 2	0	0	0	-1/4	1	-10

$$\mathbf{x} = (8, 2, 0, 0), z = 10$$

nonbasic variables typically have reduced costs $\neq 0$. Here x_3 has r.c. = 0. Let's make it enter the basis

- x_3 enters, x_2 leaves

	x_1	x_2	x_3	x_4	$-z$	b
I' = 5I	0	5	1	-5/4	0	10
II' = II + I' / 5	1	1	0	4	0	10
III' = III	0	0	0	-1/4	1	-10

$$\mathbf{x} = (10, 0, 10, 0), z = 10$$

There are 2 optimal solutions $\rightsquigarrow$ all their convex combinations are optimal solutions (from the proof of the fundamental theorem of LP) $\rightsquigarrow$

$$\mathbf{x} = \sum_{i=1}^2 \alpha_i \mathbf{x}_i$$

$$\alpha_i \geq 0$$

$$\sum_{i=1}^2 \alpha_i = 1$$

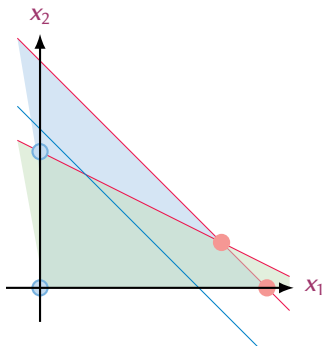
$$\mathbf{x}_1^T = [8, 2, 0, 0]$$

$$\mathbf{x}_2^T = [10, 0, 10, 0]$$

$$\alpha_1 = \alpha$$

$$\alpha_2 = 1 - \alpha$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \alpha \begin{bmatrix} 8 \\ 2 \\ 0 \\ 0 \end{bmatrix} + (1 - \alpha) \begin{bmatrix} 10 \\ 0 \\ 10 \\ 0 \end{bmatrix}$$



$$x_1 = 8\alpha + 10(1 - \alpha)$$

$$x_2 = 2\alpha$$

$$x_3 = 10(1 - \alpha)$$

$$x_4 = 0$$

Degeneracy

$$\begin{aligned}
 \max \quad & x_2 \\
 -x_1 + x_2 \leq & 0 \\
 x_1 \leq & 2 \\
 x_1, x_2 \geq & 0
 \end{aligned}$$

- Initial tableau

	x1	x2	x3	x4	-z	b
x3	-1	1	1	0	0	0
x4	1	0	0	1	0	2
	0	1	0	0	1	0

$b_i = 0$ (one basic var. is zero) might lead to cycling

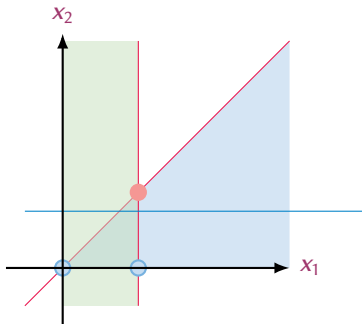
- degenerate pivot step: not improving, the entering variable stays at zero

	x1	x2	x3	x4	-z	b
	-1	1	1	0	0	0
	1	0	0	1	0	2
	1	0	-1	0	1	0

- now nondegenerate:

	x_1	x_2	x_3	x_4	$-z$	b
	0	1	0	1	0	2
	1	0	0	1	0	2
	0	0	-1	-1	1	-2

$$x_1 = 2, x_2 = 2, z = 2$$



$\geq n + 1$ constraints meet at a vertex

Def: An **improving variable** is one with positive reduced cost

Def: A **degenerate iteration** is one in which the objective function does not increase.

Def: The simplex method **cycles** if the same tableau appears in two iterations.

Degenerate conditions may appear often in practice but cycling is rare.
(see compendium for the smallest possible example)

Theorem

If the simplex fails to terminate, then it must cycle.

Proof:

- there is a finite number of basis and simplex chooses to always increase the objective value
- hence the only situation for not terminating is that a basis appears again and iterations in between are degenerate. Two tableaux with the same basis are the same (related to uniqueness of basic solutions)

Pivot Rules

Some pivot rules can prevent the occurrence of cycling altogether.

So far we chose an **arbitrary improving variable** to enter. Rules for breaking ties in selecting **entering** improving variables (more important than selecting leaving variables):

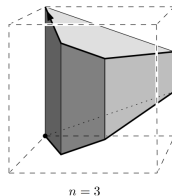
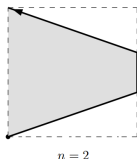
- **Largest Coefficient**: the improving var with largest coefficient in last row of the tableau.
Original Dantzig's rule, can cycle
- **Largest increase**: absolute improvement: $\operatorname{argmax}_j \{c_j \theta_j\}$
computationally more costly
- **Steepest edge**: the improving var that if entering in the basis moves the current basic feasible sol in a direction closest to the direction of the vector $\mathbf{c}$ (ie, maximizes the cosine of the angle between the two vectors):

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta \quad \Rightarrow \quad \max_{\mathbf{x}_{\text{new}}} \frac{\mathbf{c}^T (\mathbf{x}_{\text{new}} - \mathbf{x}_{\text{old}})}{\|\cancel{\mathbf{c}}\| \|\mathbf{x}_{\text{new}} - \mathbf{x}_{\text{old}}\|}$$

- **Bland's rule (smallest-subscript rule)** chooses the improving var with the lowest index and, if there are more than one leaving variable, the one with the lowest index.
Prevents cycling but is slow (no smart choice for entering variable)
- **Random edge** rule selects var uniformly at random among the improving ones and the leaving ones among those selected by the ratio test
- Charnes' **perturbation method**: perturbs values of b_i terms to avoid $b_i = 0$, which must occur for cycling. To avoid cancellations: $0 < \epsilon_m \ll \epsilon_{m-1} \ll \dots \ll \epsilon_1 \ll 1$
It affects the choice of the leaving variable.
It is equivalent to Dantzig-Orden-Wolfe's lexicographic formulation (imposes a strict ordering on the bases encountered by the simplex algorithm)
It prevents cycling.

Efficiency of Simplex Method

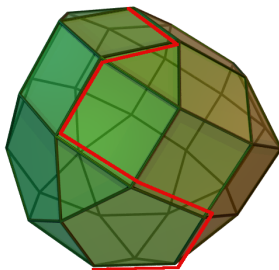
- Trying all points is $\approx 4^m$
- In practice between $2m$ and $3m$ iterations
- Klee and Minty 1978 constructed an example that requires $2^n - 1$ iterations in $\mathbb{R}^n$:



- random shuffle of indexes + lowest index for entering + lexicographic for leaving: expected iterations $< e^{C\sqrt{n \ln n}}$

Efficiency of Simplex Method

- unknown if there exists a pivot rule that leads to polynomial time.
- Clairvoyant's rule: shortest possible sequence of steps
Hirsh conjecture $O(n - d)$ for an n -facet polytope in d -dimensional Euclidean space but best known $n^{1+\ln n}$



Smoothed Complexity

Smoothed complexity: slight random perturbations of worst-case inputs

D. Spielman and S. Teng (2001), *Smoothed analysis of algorithms: why the simplex algorithm usually takes polynomial time*

Let $\bar{A} \in \mathbb{R}^{m \times n}$ have rows of norm ≤ 1 , $\bar{\mathbf{b}} \in [-1, 1]^m$, $\mathbf{c} \in \mathbb{R}^n$

Let $\hat{A}, \hat{\mathbf{b}}$ have i.i.d. $N(0, \sigma^2)$ entries

Theorem

$$\max_{\bar{A}, \bar{\mathbf{b}}, \mathbf{c}} \mathbb{E}_{\hat{A}, \hat{\mathbf{b}}} \left[\begin{array}{l} \text{time to solve} \\ \text{maximize } \mathbf{c}^T \mathbf{x} \\ \text{s.t. } (\bar{A} + \hat{A})\mathbf{x} \leq \bar{\mathbf{b}} + \hat{\mathbf{b}} \end{array} \right] \leq \text{poly}(m, n, \sigma^{-1}) = O(\max\{n^5 \log^2 m, n^9 \log^4 n, n^3 \sigma^{-4}\})$$

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Initial Infeasibility

$$\begin{aligned}\max \quad & x_1 - x_2 \\ & x_1 + x_2 \leq 2 \\ & 2x_1 + 2x_2 \geq 5 \\ & x_1, x_2 \geq 0\end{aligned}$$

$$\begin{aligned}\max \quad & x_1 - x_2 \\ & x_1 + x_2 + x_3 = 2 \\ & -2x_1 - 2x_2 + x_4 = -5 \\ & x_1, x_2, x_3, x_4 \geq 0\end{aligned}$$

- Initial tableau

	x1	x2	x3	x4	-z	b
x3	1	1	1	0	0	2
x4	-2	-2	0	1	0	-5
	1	-1	0	0	1	0

~> we do not have an initial basic feasible solution!!

In general finding any feasible solution is difficult as finding an optimal solution, otherwise we could do binary search

Auxiliary Problem (I Phase of Simplex)

We introduce auxiliary variables:

$$\begin{aligned}
 w^* &= \max -x_5 \equiv \min x_5 \\
 x_1 + x_2 + x_3 &= 2 \\
 2x_1 + 2x_2 - x_4 + x_5 &= 5 \\
 x_1, x_2, x_3, x_4, x_5 &\geq 0
 \end{aligned}$$

if $w^* = 0$ then $x_5 = 0$ and the two problems are equivalent

if $w^* > 0$ then not possible to set x_5 to zero.

- Initial tableau

	x1	x2	x3	x4	x5	-z	-w	b
	1	1	1	0	0	0	0	2
	2	2	0	-1	1	0	0	5
z	1	-1	0	0	0	1	0	0
w	0	0	0	0	-1	0	1	0

Keep z always in basis

- we reach a canonical form simply by letting x_5 enter the basis:

	x_1	x_2	x_3	x_4	x_5	$-z$	$-w$	b
	1	1	1	0	0	0	0	2
	2	2	0	-1	1	0	0	5
z	1	-1	0	0	0	1	0	0
IV+II	2	2	0	-1	0	0	1	5

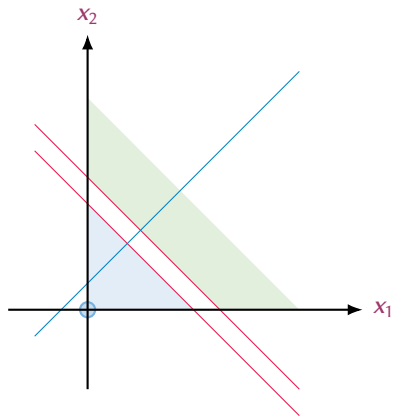
now we have a basic feasible solution!

- x_1 enters, x_3 leaves

	x_1	x_2	x_3	x_4	x_5	$-z$	$-w$	b
	1	1	1	0	0	0	0	2
II-2I'	0	0	-2	-1	1	0	0	1
III-I'	0	-2	-1	0	0	1	0	-2
IV-2I'	0	0	-2	-1	0	0	1	1

$w^* = -1$ then no solution with $x_5 = 0$ exists then no feasible solution to initial problem

$$\begin{array}{llll} \max & x_1 & - & x_2 \\ & x_1 & + & x_2 \leq 2 \\ & 2x_1 & + & 2x_2 \geq 5 \\ & & & x_1, x_2 \geq 0 \end{array}$$



Initial Infeasibility - Another Example

$$\begin{aligned} \max \quad & x_1 - x_2 \\ & x_1 + x_2 \leq 2 \\ & 2x_1 + 2x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$$\begin{aligned} \max \quad & x_1 - x_2 \\ & x_1 + x_2 + x_3 = 2 \\ & 2x_1 + 2x_2 - x_4 = 2 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Auxiliary problem (I phase):

$$\begin{aligned} w = \max \quad & -x_5 \equiv \min x_5 \\ & x_1 + x_2 + x_3 = 2 \\ & 2x_1 + 2x_2 - x_4 + x_5 = 2 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned}$$

- Initial tableau

	x1	x2	x3	x4	x5	-z	-w	b
	1	1	1	0	0	0	0	2
	2	2	0	-1	1	0	0	2
z	1	-1	0	0	0	1	0	0
w	0	0	0	0	-1	0	1	0

→ we do not have an initial basic feasible solution.

- set in canonical form:

	x1	x2	x3	x4	x5	-z	-w	b
	1	1	1	0	0	0	0	2
	2	2	0	-1	1	0	0	2
z	1	-1	0	0	0	1	0	0
IV+II	2	2	0	-1	0	0	1	2

- x_1 enters, x_5 leaves

	x1	x2	x3	x4	x5	-z	-w	b
	0	0	1	1/2	-1/2	0	0	1
	1	1	0	-1/2	1/2	0	0	1
z	0	-2	0	1/2	-1/2	1	0	-1
w	0	0	0	0	-1	0	1	0

$w^* = 0$ hence $x_5 = 0$ we have a starting feasible solution for the initial problem.

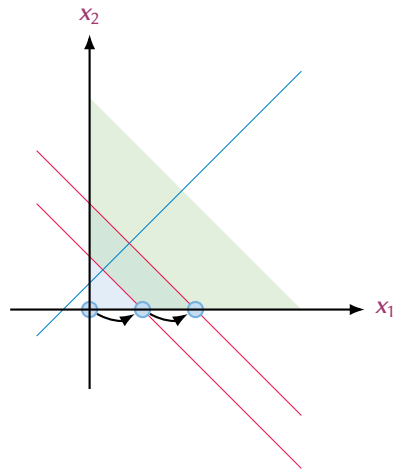
- (II phase) We keep only what we need:

	x1	x2	x3	x4	-z	b
	0	0	1	1/2	0	1
	1	1	0	-1/2	0	1
	z	0	-2	0	1/2	1

- | | x1 | x2 | x3 | x4 | -z | b |
|--|----|----|----|----|----|---|
| | | | | | | |
| | 0 | 0 | 2 | 1 | 0 | 2 |
| | 1 | 1 | 1 | 0 | 0 | 2 |
| | z | 0 | -2 | -1 | 0 | 1 |

Optimal solution: $x_1 = 2, x_2 = 0, x_3 = 0, x_4 = 2, z = 2$.

$$\begin{aligned} \max \quad & x_1 - x_2 \\ & x_1 + x_2 \leq 2 \\ & 2x_1 + 2x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$



In Dictionary Form

$$\begin{array}{rcl}
 \max & x_1 & - x_2 \\
 & x_1 + x_2 & \leq 2 \\
 & 2x_1 + 2x_2 & \geq 5 \\
 & x_1, x_2 & \geq 0
 \end{array}$$

$$\begin{array}{rcl}
 x_3 & = & 2 - x_1 - x_2 \\
 x_4 & = & -5 + 2x_1 + 2x_2 \\
 \hline
 z & = & x_1 + x_2
 \end{array}$$

sol. infeasible

We introduce corrections of infeasibility

$$\begin{array}{rcl}
 \max & -x_0 & \equiv \min x_0 \\
 & x_1 + x_2 & \leq 2 \\
 & 2x_1 + 2x_2 - x_0 & \geq 5 \\
 & x_1, x_2, x_0 & \geq 0
 \end{array}$$

$$\begin{array}{rcl}
 x_3 & = & 2 - x_1 - x_2 \\
 x_4 & = & -5 + 2x_1 + 2x_2 + x_0 \\
 \hline
 z & = & - x_0
 \end{array}$$

It is still infeasible but it can be made feasible by letting x_0 enter the basis
which variable should leave?

the most infeasible: the var with the b term whose negative value has the largest magnitude

Simplex: Exception Handling, Summary

Solution of an LP problem:

- a. $F \neq \emptyset$ and $\nexists$ solution
- b. $F \neq \emptyset$ and $\exists$ solution
 - i) **one solution**
 - ii) infinite solutions
- c. $F = \emptyset$

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- 1. Unboundedness
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Dual Problem

Dual variables $\mathbf{y}$ in one-to-one correspondence with the constraints:

Primal problem:

$$\begin{aligned}\max \quad & z = \mathbf{c}^T \mathbf{x} \\ & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{aligned}$$

Dual Problem:

$$\begin{aligned}\min \quad & w = \mathbf{b}^T \mathbf{y} \\ & A^T \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq \mathbf{0}\end{aligned}$$

Bounding approach

$$\begin{aligned} z^* = \max \quad & 4x_1 + x_2 + 3x_3 \\ & x_1 + 4x_2 \leq 1 \\ & 3x_1 + x_2 + x_3 \leq 3 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

a feasible solution is a **lower bound** but how good?

By tentatives:

$$(x_1, x_2, x_3) = (1, 0, 0) \rightsquigarrow z^* \geq 4$$

$$(x_1, x_2, x_3) = (0, 0, 3) \rightsquigarrow z^* \geq 9$$

What about **upper bounds**?

$$\begin{array}{rcl} & 2 \cdot (x_1 + 4x_2) & \leq 2 \cdot 1 \\ & + 3 \cdot (3x_1 + x_2 + x_3) & \leq 3 \cdot 3 \\ \hline 4x_1 + x_2 + 3x_3 & \leq & 11x_1 + 11x_2 + 3x_3 \leq 11 \end{array}$$
$$\mathbf{c}^T \mathbf{x} \leq \mathbf{y}^T A \mathbf{x} \leq \mathbf{y}^T \mathbf{b}$$

Hence $z^* \leq 11$. Is this the best upper bound we can find?

multipliers $y_1, y_2 \geq 0$ that preserve sign of inequality

$$\begin{array}{rcl} y_1 \cdot (x_1 + 4x_2) & \leq & y_1(1) \\ y_2 \cdot (3x_1 + x_2 + x_3) & \leq & y_2(3) \\ \hline (y_1 + 3y_2)x_1 + (4y_1 + y_2)x_2 + y_2x_3 & \leq & y_1 + 3y_2 \end{array}$$

Coefficients

$$\begin{array}{rcl} y_1 + 3y_2 & \geq & 4 \\ 4y_1 + y_2 & \geq & 1 \\ y_2 & \geq & 3 \end{array}$$

$z = 4x_1 + x_2 + 3x_3 \leq (y_1 + 3y_2)x_1 + (4y_1 + y_2)x_2 + y_2x_3 \leq y_1 + 3y_2$
then to attain the best upper bound:

$$\begin{array}{rcl} \min & y_1 + 3y_2 \\ & y_1 + 3y_2 \geq 4 \\ & 4y_1 + y_2 \geq 1 \\ & y_2 \geq 3 \\ & y_1, y_2 \geq 0 \end{array}$$

Multipliers Approach

$$\begin{array}{l} \pi_1 \\ \vdots \\ \pi_m \\ \pi_{m+1} \end{array} \left[\begin{array}{cccc|cccc|c|c} a_{11} & a_{12} & \dots & a_{1n} & a_{1,n+1} & a_{1,n+2} & \dots & a_{1,m+n} & 0 & b_1 \\ \vdots & \ddots & & & & & & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} & a_{m,n+1} & a_{m,n+2} & \dots & a_{m,m+n} & 0 & b_m \\ \hline c_1 & c_2 & \dots & c_n & 0 & 0 & \dots & 0 & 1 & 0 \end{array} \right]$$

Working columnwise, since at optimum $\bar{c}_k \leq 0$ for all $k = 1, \dots, n + m$:

$$\left\{ \begin{array}{l} \pi_1 a_{11} + \pi_2 a_{21} \dots + \pi_m a_{m1} + \pi_{m+1} c_1 \leq 0 \\ \vdots \quad \ddots \hspace{6em} \vdots \\ \pi_1 a_{1n} + \pi_2 a_{2n} \dots + \pi_m a_{mn} + \pi_{m+1} c_n \leq 0 \\ - \text{-----} \\ \pi_1 a_{1,n+1}, \quad \pi_2 a_{2,n+1}, \quad \dots \quad \pi_m a_{m,n+1} \leq 0 \\ \vdots \hspace{8em} \vdots \hspace{8em} \vdots \hspace{8em} \vdots \hspace{8em} \vdots \hspace{8em} \vdots \\ \pi_1 a_{1,n+m}, \quad \pi_2 a_{2,n+m}, \quad \dots \quad \pi_m a_{m,n+m} \leq 0 \\ - \text{-----} \\ \hspace{27em} \pi_{m+1} = 1 \\ - \text{-----} \\ \pi_1 b_1 + \pi_2 b_2 \dots + \pi_m b_m (\leq 0) \end{array} \right.$$

(from the last row we have also $z = -\pi \mathbf{b}$)

$$\begin{array}{rcl}
-z = & \pi_1 b_1 & + \pi_2 b_2 \dots + \pi_m b_m \\
& \pi_1 a_{11} & + \pi_2 a_{21} \dots + \pi_m a_{m1} \leq -c_1 \\
& \vdots & \ddots \qquad \qquad \qquad \vdots \\
& \pi_1 a_{1n} & + \pi_2 a_{2n} \dots + \pi_m a_{mn} \leq -c_n \\
& & \pi_1, \pi_2, \dots, \pi_m \leq 0
\end{array}$$

$$\mathbf{y} = -\boldsymbol{\pi}$$

$$\begin{array}{rcl}
-z = & (-y_1 b_1) & + (-y_2 b_2) \dots + (-y_m b_m) \\
& (-y_1 a_{11}) & + (-y_2 a_{21}) \dots + (-y_m a_{m1}) \leq -c_1 \\
& \vdots & \ddots \qquad \qquad \qquad \vdots \\
& (-y_1 a_{1n}) & + (-y_2 a_{2n}) \dots + (-y_m a_{mn}) \leq -c_n \\
& & -y_1, -y_2, \dots, -y_m \leq 0
\end{array}$$

as we will see $\mathbf{b}^T \mathbf{y} \geq \mathbf{c}^T \mathbf{x}$, hence it is more interesting to minimize. It yields:

$$\begin{array}{ll}
\min & \mathbf{b}^T \mathbf{y} \\
& A^T \mathbf{y} \geq \mathbf{c} \\
& \mathbf{y} \geq \mathbf{0}
\end{array}$$

Example

$$\begin{array}{ll}\max & 6x_1 + 8x_2 \\ & 5x_1 + 10x_2 \leq 60 \\ & 4x_1 + 4x_2 \leq 40 \\ & x_1, x_2 \geq 0\end{array}$$

$$\left\{ \begin{array}{lll} 5\pi_1 + 4\pi_2 + 6\pi_3 \leq 0 \\ 10\pi_1 + 4\pi_2 + 8\pi_3 \leq 0 \\ 1\pi_1 + 0\pi_2 + 0\pi_3 \leq 0 \\ 0\pi_1 + 1\pi_2 + 0\pi_3 \leq 0 \\ 0\pi_1 + 0\pi_2 + 1\pi_3 = 1 \\ 60\pi_1 + 40\pi_2 \end{array} \right.$$

$$y_1 = -\pi_1 \geq 0$$

$$y_2 = -\pi_2 \geq 0$$

Duality Recipe

	Primal linear program	Dual linear program
Variables	$x_1, x_2, \dots, x_n$	$y_1, y_2, \dots, y_m$
Matrix	A	A^T
Right-hand side	$\mathbf{b}$	$\mathbf{c}$
Objective function	$\max \mathbf{c}^T \mathbf{x}$	$\min \mathbf{b}^T \mathbf{y}$
Constraints	i th constraint has $\leq$ $\geq$ $=$	$y_i \geq 0$ $y_i \leq 0$ $y_i \in \mathbb{R}$
	$x_j \geq 0$ $x_j \leq 0$ $x_j \in \mathbb{R}$	j th constraint has $\geq$ $\leq$ $=$

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Symmetry

The dual of the dual is the primal:

Primal Problem:

$$\begin{aligned}\max \quad & z = c^T x \\ & Ax \leq b \\ & x \geq 0\end{aligned}$$

Let's put the dual in the standard form

Dual Problem:

$$\begin{aligned}\min \quad & b^T y \equiv -\max -b^T y \\ & -A^T y \leq -c \\ & y \geq 0\end{aligned}$$

Dual Problem:

$$\begin{aligned}\min \quad & w = b^T y \\ & A^T y \geq c \\ & y \geq 0\end{aligned}$$

Dual of Dual:

$$\begin{aligned}-\min \quad & -c^T x \\ & -Ax \geq -b \\ & x \geq 0\end{aligned}$$

Weak Duality Theorem

As we saw the dual produces upper bounds. This is true in general:

Theorem (Weak Duality Theorem)

Given:

$$(P) \max\{\mathbf{c}^T \mathbf{x} \mid A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$$

$$(D) \min\{\mathbf{b}^T \mathbf{y} \mid A^T \mathbf{y} \geq \mathbf{c}, \mathbf{y} \geq \mathbf{0}\}$$

for any feasible solution $\mathbf{x}$ of (P) and any feasible solution $\mathbf{y}$ of (D):

$$\mathbf{c}^T \mathbf{x} \leq \mathbf{b}^T \mathbf{y}$$

Proof:

From (D) $c_j \leq \sum_{i=1}^m y_i a_{ij} \forall j$ and from (P) $\sum_{j=1}^n a_{ij} x_j \leq b_i \forall i$

From (D) $y_i \geq 0$ and from (P) $x_j \geq 0$

$$\sum_{j=1}^n c_j x_j \leq \sum_{j=1}^n \left(\sum_{i=1}^m y_i a_{ij} \right) x_j = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j \right) y_i \leq \sum_{i=1}^m b_i y_i$$

Strong Duality Theorem

Due to Von Neumann and Dantzig 1947 and Gale, Kuhn and Tucker 1951.

Theorem (Strong Duality Theorem)

Given:

$$(P) \max\{\mathbf{c}^T \mathbf{x} \mid A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$$

$$(D) \min\{\mathbf{b}^T \mathbf{y} \mid A^T \mathbf{y} \geq \mathbf{c}, \mathbf{y} \geq \mathbf{0}\}$$

exactly one of the following occurs:

1. *(P) and (D) are both infeasible*
2. *(P) is unbounded and (D) is infeasible*
3. *(P) is infeasible and (D) is unbounded*
4. *(P) has feasible solution, then let an optimal be: $\mathbf{x}^* = [x_1^*, \dots, x_n^*]$
(D) has feasible solution, then let an optimal be: $\mathbf{y}^* = [y_1^*, \dots, y_m^*]$, then:*

$$\mathbf{c}^T \mathbf{x}^* = \mathbf{b}^T \mathbf{y}^*$$

Proof:

- all other combinations of 3 possibilities (Optimal, Infeasible, Unbounded) for (P) and 3 for (D) are ruled out by weak duality theorem.
- we use the simplex method. (Other proofs independent of the simplex method exist, eg, Farkas Lemma and convex polyhedral analysis)
- The last row of the final tableau will give us that for every $\mathbf{x}$:

$$\begin{aligned} z &= z^* + \sum_{k=1}^{n+m} \bar{c}_k x_k = z^* + \sum_{j=1}^n \bar{c}_j x_j + \sum_{i=1}^m \bar{c}_{n+i} x_{n+i} \\ &= z^* + \bar{\mathbf{c}}_B \mathbf{x}_B + \bar{\mathbf{c}}_N \mathbf{x}_N \end{aligned} \quad (*)$$

In addition, $z^* = \sum_{j=1}^n c_j x_j^*$ (c_j , original values) because optimal value

- We define $y_i^* = -\bar{c}_{n+i}, \quad i = 1, 2, \dots, m$

- We claim that $(y_1^*, y_2^*, \dots, y_m^*)$ is a dual feasible solution satisfying $\mathbf{c}^T \mathbf{x}^* = \mathbf{b}^T \mathbf{y}^*$.

- Let's verify the claim:

We substitute in (*): i) $z = \sum_{j=1}^n c_j x_j$; ii) $\bar{c}_{n+i} = -y_i^*$; and iii) $x_{n+i} = b_i - \sum_{j=1}^n a_{ij} x_j$ for $i = 1, 2, \dots, m$ ($n+i$ are the slack variables)

$$\begin{aligned} \sum_{j=1}^n c_j x_j &= z^* + \sum_{j=1}^n \bar{c}_j x_j - \sum_{i=1}^m y_i^* \left(b_i - \sum_{j=1}^n a_{ij} x_j \right) \\ &= \left(z^* - \sum_{i=1}^m y_i^* b_i \right) + \sum_{j=1}^n \left(\bar{c}_j + \sum_{i=1}^m a_{ij} y_i^* \right) x_j \end{aligned}$$

This must hold for every $(x_1, x_2, \dots, x_n)$ hence:

$$\begin{aligned} z^* &= \sum_{i=1}^m b_i y_i^* & \implies y^* \text{ satisfies } c^T x^* &= b^T y^* \\ c_j &= \bar{c}_j + \sum_{i=1}^m a_{ij} y_i^*, j = 1, 2, \dots, n \end{aligned}$$

Since $\bar{c}_k \leq 0$ for every $k = 1, 2, \dots, n + m$:

$$\bar{c}_j \leq 0 \rightsquigarrow c_j - \sum_{i=1}^m y_i^* a_{ij} \leq 0 \rightsquigarrow$$

$$\sum_{i=1}^m y_i^* a_{ij} \geq c_j \quad j = 1, 2, \dots, n$$

$$\bar{c}_{n+i} \leq 0 \rightsquigarrow y_i^* = -\bar{c}_{n+i} \geq 0,$$

$$i = 1, 2, \dots, m$$

$\Rightarrow y^*$ is also dual feasible solution

Complementary Slackness Theorem

Theorem (Complementary Slackness)

A feasible solution $\mathbf{x}^*$ for (P)

A feasible solution $\mathbf{y}^*$ for (D)

Necessary and sufficient conditions for optimality of both:

$$\left(c_j - \sum_{i=1}^m y_i^* a_{ij} \right) x_j^* = 0, \quad j = 1, \dots, n$$

If $x_j^* \neq 0$ then $\sum y_i^* a_{ij} = c_j$ (no surplus)

If $\sum y_i^* a_{ij} > c_j$ then $x_j^* = 0$

Proof:

$$z^* = \mathbf{c}^T \mathbf{x}^* \leq \mathbf{y}^* \mathbf{A} \mathbf{x}^* \leq \mathbf{b}^T \mathbf{y}^* = w^*$$

Hence from strong duality theorem:

$$\mathbf{c} \mathbf{x}^* - \mathbf{y}^* \mathbf{A} \mathbf{x}^* = 0$$

In scalars

$$\sum_{j=1}^n \underbrace{\left(c_j - \sum_{i=1}^m y_i^* a_{ij} \right)}_{\leq 0} \underbrace{x_j^*}_{\geq 0} = 0$$

Hence each term must be $= 0$

Proof in scalar form:

$$c_j x_j^* \leq \left(\sum_{i=1}^m a_{ij} y_i^* \right) x_j^* \quad j = 1, 2, \dots, n \quad \text{from feasibility in D}$$

$$\left(\sum_{j=1}^n a_{ij} x_j^* \right) y_i^* \leq b_i y_i^* \quad i = 1, 2, \dots, m \quad \text{from feasibility in P}$$

Summing in j and in i :

$$\sum_{j=1}^n c_j x_j^* \leq \sum_{j=1}^n \left(\sum_{i=1}^m a_{ij} y_i^* \right) x_j^* = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j^* \right) y_i^* \leq \sum_{i=1}^m b_i y_i^*$$

For the strong duality theorem the left hand side is equal to the right hand side and hence all inequalities become equalities.

$$\sum_{j=1}^n \underbrace{\left(c_j - \sum_{i=1}^m y_i^* a_{ij} \right)}_{\leq 0} \underbrace{x_j^*}_{\geq 0} = 0$$

Economic Interpretation of Duality Theory

$$\begin{aligned}
 \max \quad & 5x_0 + 6x_1 + 8x_2 \\
 & 6x_0 + 5x_1 + 10x_2 \leq 60 \\
 & 8x_0 + 4x_1 + 4x_2 \leq 40 \\
 & 4x_0 + 5x_1 + 6x_2 \leq 50 \\
 & x_0, x_1, x_2 \geq 0
 \end{aligned}$$

final tableau:

$$\begin{array}{ccccccc|c}
 x_0 & x_1 & x_2 & s_1 & s_2 & s_3 & -z & b \\
 \hline
 & 0 & 1 & & 0 & & & 5/2 \\
 & 1 & 0 & & 0 & & & 7 \\
 & 0 & 0 & & 1 & & & 2 \\
 \hline
 -1/5 & 0 & 0 & -1/5 & 0 & -1 & & -62
 \end{array}$$

- What is the maximum number of products that can be produced? at most m
- What are the values of the variables, the reduced costs, the dual variables, and the shadow (or marginal) prices?
- What is the value of additional manpower capacity (1st constr.)? In +1 out +1/5
- If one slack variable > 0 then overcapacity: $s_2 = 2$ then the second constraint is not tight
- How much more expensive would a non-selected product need to be? look at reduced costs:

$$c_j + \pi a_j > 0$$

Economic Interpretation of Duality Theory

Game: Suppose two economic operators:

- P owns the factory and produces goods
- D is in the market buying and selling raw material and resources
- D asks P to close and sell him/her all resources
- P considers if the offer is convenient
- D wants to spend the least possible
- y are prices that D offers for the resources
- $\sum y_i b_i$ is the amount D has to pay to have all resources of P
- $\sum y_i a_{ij} \geq c_j$ the total value to make j is greater or eq. than the price per unit of product j
- P either sells all resources $\sum y_i a_{ij}$ or produces product j (c_j)
- without $\geq$ there would not be negotiation because P would be better off producing and selling
- ▶ at optimality the situation is indifferent (strong th.)
- ▶ resource 2 that was not totally utilized in the primal has been given value 0 in the dual.
(complementary slackness th.) Plausible, since we do not use all the resource, likely to place not so much value on it.
- ▶ for product 0 $\sum y_i a_{ij} > c_j$ hence not profitable producing it. (complementary slackness th.)

Duality - Summary

- Derivation:
 - Economic Interpretation
 - Bounding Approach
 - Multiplier Approach
 - Recipe
 - Lagrangian Multipliers Approach (next time)
- Theory:
 - Symmetry
 - Weak Duality Theorem
 - Strong Duality Theorem
 - Complementary Slackness Theorem

Summary

- Derivation:
 1. economic interpretation
 2. bounding
 3. multipliers
 4. recipe
 5. Lagrangian
- Theory:
 - Symmetry
 - Weak duality theorem
 - Strong duality theorem
 - Complementary slackness theorem
- Economic interpretation
- Dual Simplex
- Sensitivity Analysis

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Lagrangian Duality

Relaxation: if a problem is hard to solve then find an easier problem resembling the original one that provides information in terms of bounds. Then, search for the strongest bounds.

$$\begin{aligned} \min \quad & 13x_1 + 6x_2 + 4x_3 + 12x_4 \\ & 2x_1 + 3x_2 + 4x_3 + 5x_4 = 7 \\ & 3x_1 + \quad + 2x_3 + 4x_4 = 2 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

We wish to reduce to a problem easier to solve, ie:

$$\begin{aligned} \min \quad & c_1x_1 + c_2x_2 + \dots + c_nx_n \\ & x_1, x_2, \dots, x_n \geq 0 \end{aligned}$$

solvable by inspection: if $c_j < 0$ then $x_j = +\infty$, if $c_j \geq 0$ then $x_j = 0$.

Measure of violation of the constraints:

$$\begin{aligned} 7 - (2x_1 + 3x_2 + 4x_3 + 5x_4) \\ 2 - (3x_1 + \quad + 2x_3 + 4x_4) \end{aligned}$$

We relax these measures in obj. function with **Lagrangian multipliers** y_1, y_2 .

We obtain a family of problems:

$$PR(y_1, y_2) = \min_{x_1, x_2, x_3, x_4 \geq 0} \left\{ \begin{array}{l} 13x_1 + 6x_2 + 4x_3 + 12x_4 \\ + y_1(7 - 2x_1 - 3x_2 - 4x_3 - 5x_4) \\ + y_2(2 - 3x_1 - 2x_3 - 4x_4) \end{array} \right\}$$

1. for all $y_1, y_2 \in \mathbb{R} : \text{opt}(PR(y_1, y_2)) \leq \text{opt}(P)$
2. $\max_{y_1, y_2 \in \mathbb{R}} \{\text{opt}(PR(y_1, y_2))\} \leq \text{opt}(P)$

PR is easy to solve.

(It can be also seen as a proof of the weak duality theorem)

$$PR(y_1, y_2) = \min_{x_1, x_2, x_3, x_4 \geq 0} \left\{ \begin{array}{l} (13 - 2y_1 - 3y_2) x_1 \\ + (6 - 3y_1) x_2 \\ + (4 - 4y_1 - 2y_2) x_3 \\ + (12 - 5y_1 - 4y_2) x_4 \\ + 7y_1 + 2y_2 \end{array} \right\}$$

if coefficient of x is < 0 then bound is $-\infty$ then LB is useless

$$\begin{aligned} (13 - 2y_1 - 3y_2) &\geq 0 \\ (6 - 3y_1) &\geq 0 \\ (4 - 4y_1 - 2y_2) &\geq 0 \\ (12 - 5y_1 - 4y_2) &\geq 0 \end{aligned}$$

If they all hold then we are left with $7y_1 + 2y_2$ because all go to 0.

$$\begin{aligned} \max \quad & 7y_1 + 2y_2 \\ & 2y_1 + 3y_2 \leq 13 \\ & 3y_1 \leq 6 \\ & 4y_1 + 2y_2 \leq 4 \\ & 5y_1 + 4y_2 \leq 12 \end{aligned}$$

General Formulation

$$\begin{array}{ll} \min & z = \mathbf{c}^T \mathbf{x} & \mathbf{c} \in \mathbb{R}^n \\ & A\mathbf{x} = \mathbf{b} & A \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m \\ & \mathbf{x} \geq \mathbf{0} & \mathbf{x} \in \mathbb{R}^n \end{array}$$

$$\max_{\mathbf{y} \in \mathbb{R}^m} \left\{ \min_{\mathbf{x} \in \mathbb{R}_+^n} \{ \mathbf{c}^T \mathbf{x} + \mathbf{y}^T (\mathbf{b} - A\mathbf{x}) \} \right\}$$

$$\max_{\mathbf{y} \in \mathbb{R}^m} \left\{ \min_{\mathbf{x} \in \mathbb{R}_+^n} \{ (\mathbf{c}^T - \mathbf{y}^T A) \mathbf{x} + \mathbf{y}^T \mathbf{b} \} \right\}$$

$$\begin{array}{ll} \max & \mathbf{b}^T \mathbf{y} \\ & A^T \mathbf{y} \leq \mathbf{c} \\ & \mathbf{y} \in \mathbb{R}^m \end{array}$$

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Dual Simplex

- Dual simplex (Lemke, 1954): apply the simplex method to the dual problem and observe what happens in the primal tableau:

$$\begin{aligned}\max\{\mathbf{c}^T \mathbf{x} \mid A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}\} &= \min\{\mathbf{b}^T \mathbf{y} \mid A^T \mathbf{y} \geq \mathbf{c}^T, \mathbf{y} \geq \mathbf{0}\} \\ &= -\max\{-\mathbf{b}^T \mathbf{y} \mid -A^T \mathbf{y} \leq -\mathbf{c}^T, \mathbf{y} \geq \mathbf{0}\}\end{aligned}$$

- We obtain a new algorithm for the primal problem: the **dual simplex**
It corresponds to the primal simplex applied to the dual
- We obtain also another way to handle infeasibility

Primal Simplex on Dual Problem

Example: Primal feasible start

Primal:

$$\begin{aligned} \max \quad & 4x_1 + x_2 + 3x_3 \\ & x_1 + 4x_2 \leq 1 \\ & 3x_1 + x_2 + x_3 \leq 3 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

- Initial tableau

x1	x2	x3	w1	w2	-z	b
1	4	0	1	0	0	1
3	1	1	0	1	0	3
4	1	3	0	0	1	0

- primal feasible start

Dual:

$$\begin{aligned} \min \quad & y_1 + 3y_2 \\ & y_1 + 3y_2 \geq 4 \\ & 4y_1 + y_2 \geq 1 \\ & y_2 \geq 3 \\ & y_1, y_2 \geq 0 \end{aligned}$$

- Initial tableau ($\min by \equiv -\max -by$)

y1	y2	z1	z2	z3	-p	b
-1	-3	1	0	0	0	-4
-4	-1	0	0	1	0	-1
0	-1	0	0	1	0	-3
-1	-3	0	0	0	1	0

- dual infeasible start

Primal Simplex on Dual Problem

Example: Primal infeasible start

Primal:

$$\begin{aligned}
 \max \quad & -x_1 - x_2 \\
 & -2x_1 - x_2 \leq 4 \\
 & -2x_1 + 4x_2 \leq -8 \\
 & -x_1 + 3x_2 \leq -7 \\
 & x_1, x_2 \geq 0
 \end{aligned}$$

- Initial tableau

	x1	x2	w1	w2	w3	-z	b
	-2	-1	1	0	0	0	4
	-2	4	0	1	0	0	-8
	-1	3	0	0	1	0	-7
	-1	-1	0	0	0	1	0

infeasible start

- x_1 enters, w_2 leaves

Dual:

$$\begin{aligned}
 \min \quad & 4y_1 - 8y_2 - 7y_3 \\
 & -2y_1 - 2y_2 - y_3 \geq -1 \\
 & -y_1 + 4y_2 + 3y_3 \geq -1 \\
 & y_1, y_2, y_3 \geq 0
 \end{aligned}$$

- Initial tableau ($\min by \equiv -\max -by$)

	y1	y2	y3	z1	z2	-p	b
	2	2	1	1	0	0	1
	1	-4	-3	0	1	0	1
	-4	8	7	0	0	1	0

feasible start (thanks to $-x_1 - x_2$)

- y_2 enters, z_1 leaves

- x_1 enters, w_2 leaves

	x_1	x_2	w_1	w_2	w_3	$-z$	b
	0	-5	1	-1	0	0	12
	1	-2	0	-0.5	0	0	4
	0	1	0	-0.5	1	0	-3
	0	-3	0	-0.5	0	1	4

- w_2 enters, w_3 leaves

	x_1	x_2	w_1	w_2	w_3	$-z$	b
	0	-7	1	0	-2	0	18
	1	-3	0	0	-1	0	7
	0	-2	0	1	-2	0	6
	0	-4	0	0	-1	1	7

(note that we kept $c_j < 0$, ie, optimality)

- y_2 enters, z_1 leaves

	y_1	y_2	y_3	z_1	z_2	$-p$	b
	1	1	0.5	0.5	0	0	0.5
	5	0	-1	2	1	0	3
	-4	0	3	-12	0	1	-4

- y_3 enters, y_2 leaves

	y_1	y_2	y_3	z_1	z_2	$-p$	b
	2	2	1	1	0	0	1
	7	2	0	3	1	0	3
	-18	-6	0	-7	0	1	-7

Dual Simplex on Primal Problem

Primal simplex on primal problem:

1. pivot > 0
2. col c_j with wrong sign
3. row: $\min \left\{ \frac{b_i}{a_{ij}} : a_{ij} > 0, i = 1, \dots, m \right\}$

Dual simplex on primal problem:

1. pivot < 0
2. row $b_i < 0$
(condition of feasibility)
3. col: $\min \left\{ \left| \frac{c_j}{a_{ij}} \right| : a_{ij} < 0, j = 1, 2, \dots, n + m \right\}$
(least worsening solution)

- Primal works with feasible solutions towards optimality
- Dual works with optimal solutions towards feasibility

Dual Simplex

1. (primal) simplex on primal problem (the one studied so far)
2. Now: dual simplex on primal problem $\equiv$ primal simplex on dual problem
(implemented as dual simplex, understood as primal simplex on dual problem)

Uses of the Dual Simplex:

- Infeasible start
Dual based Phase I algorithm (**Dual-primal algorithm**)
- The dual simplex can work better than the primal in some cases.
Eg. since running time in practice between $2m$ and $3m$, then if $m = 99$ and $n = 9$ then better the dual

Dual based Phase I

Example:

$$\begin{array}{ll}\text{maximize} & z = x_1 - x_2 \\ \text{subject to} & x_1 + x_2 \leq 2 \\ & 2x_1 + 2x_2 \geq 2 \\ & x_1, x_2 \geq 0\end{array}$$

Summary

- Derivation:
 1. bounding
 2. multipliers
 3. recipe
 4. Lagrangian
- Theory:
 - Symmetry
 - Weak duality theorem
 - Strong duality theorem
 - Complementary slackness theorem
- Dual Simplex
- Sensitivity Analysis, Economic interpretation

Outline

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 - Resource Allocation
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Sensitivity Analysis

aka Postoptimality Analysis

Instead of solving each modified problem from scratch, exploit results obtained from solving the original problem.

$$\max\{\mathbf{c}^T \mathbf{x} \mid A\mathbf{x} = \mathbf{b}, \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}\} \quad (*)$$

(I) changes to coefficients of objective function: $\max\{\tilde{\mathbf{c}}^T \mathbf{x} \mid A\mathbf{x} = \mathbf{b}, \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}\}$ (primal)
 $\mathbf{x}^*$ of (*) remains feasible hence we can restart the simplex from $\mathbf{x}^*$

(II) changes to RHS terms: $\max\{\mathbf{c}^T \mathbf{x} \mid A\mathbf{x} = \tilde{\mathbf{b}}, \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}\}$ (dual)
 $\mathbf{x}^*$ optimal feasible solution of (*)

basic sol $\bar{\mathbf{x}}$ of (II): $\bar{\mathbf{x}}_N = \mathbf{x}_N^*, A_B \bar{\mathbf{x}}_B = \tilde{\mathbf{b}} - A_N \bar{\mathbf{x}}_N$

$\bar{\mathbf{x}}$ is dual feasible and we can start the dual simplex from there. If $\tilde{\mathbf{b}}$ differs from $\mathbf{b}$ only slightly it may be we are already optimal.

(III) introduce a new variable:

(primal)

$$\begin{aligned} \max \quad & \sum_{j=1}^6 c_j x_j \\ & \sum_{j=1}^6 a_{ij} x_j = b_i, \quad i = 1, \dots, 3 \\ & l_j \leq x_j \leq u_j, \quad j = 1, \dots, 6 \\ & [x_1^*, \dots, x_6^*] \text{ feasible} \end{aligned}$$

$$\begin{aligned} \max \quad & \sum_{j=1}^7 c_j x_j \\ & \sum_{j=1}^7 a_{ij} x_j = b_i, \quad i = 1, \dots, 3 \\ & l_j \leq x_j \leq u_j, \quad j = 1, \dots, 7 \\ & [x_1^*, \dots, x_6^*, 0] \text{ feasible} \end{aligned}$$

(IV) introduce a new constraint:

(dual)

$$\begin{aligned} & \sum_{j=1}^6 a_{4j} x_j = b_4 \\ & \sum_{j=1}^6 a_{5j} x_j = b_5 \\ & l_j \leq x_j \leq u_j \quad j = 7, 8 \end{aligned}$$

$$\begin{aligned} & [x_1^*, \dots, x_6^*] \text{ optimal} \\ & [x_1^*, \dots, x_6^*, x_7^*, x_8^*] \text{ dual feasible} \\ & x_7^* = b_4 - \sum_{j=1}^6 a_{4j} x_j^* \\ & x_8^* = b_5 - \sum_{j=1}^6 a_{5j} x_j^* \end{aligned}$$

Examples

(I) Variation of reduced costs:

$$\begin{aligned} \max \quad & 6x_1 + 8x_2 \\ & 5x_1 + 10x_2 \leq 60 \\ & 4x_1 + 4x_2 \leq 40 \\ & x_1, x_2 \geq 0 \end{aligned}$$

	x_1	x_2	x_3	x_4	$-z$	b
x_3	5	10	1	0	0	60
x_4	4	4	0	1	0	40
	6	8	0	0	1	0

The last tableau gives the possibility to estimate the effect of variations

	x_1	x_2	x_3	x_4	$-z$	b
x_2	0	1	$1/5$	$-1/4$	0	2
x_1	1	0	$-1/5$	$1/2$	0	8
	0	0	$-2/5$	-1	1	-64

- For a variable in basis the perturbation goes unchanged in the red. costs. Eg:

$$\max (6 + \delta)x_1 + 8x_2 \implies \bar{c}_1 = 1(6 + \delta) - \frac{2}{5} \cdot 5 - 1 \cdot 4 = \delta$$

then need to bring in canonical form and hence δ changes the obj value.

- For a variable not in basis, if it changes the sign of the reduced cost $\implies$ worth bringing in basis $\implies$ the δ term propagates to other columns

(II) Changes in RHS terms

	x_1	x_2	x_3	x_4	$-z$	b
x_3	5	10	1	0	0	$60 + \delta$
x_4	4	4	0	1	0	$40 + \epsilon$
	6	8	0	0	1	0

	x_1	x_2	x_3	x_4	$-z$	b
x_2	0	1	$1/5$	$-1/4$	0	$2 + 1/5\delta - 1/4\epsilon$
x_1	1	0	$-1/5$	$1/2$	0	$8 - 1/5\delta + 1/2\epsilon$
	0	0	$-2/5$	-1	1	$-64 - 2/5\delta - \epsilon$

(It would be more convenient to augment the second. But let's take $\epsilon = 0$.)

If $60 + \delta \implies$ all RHS terms change and we must check feasibility

Which are the multipliers for the first row? $k_1 = \frac{1}{5}, k_2 = -\frac{1}{4}, k_3 = 0$

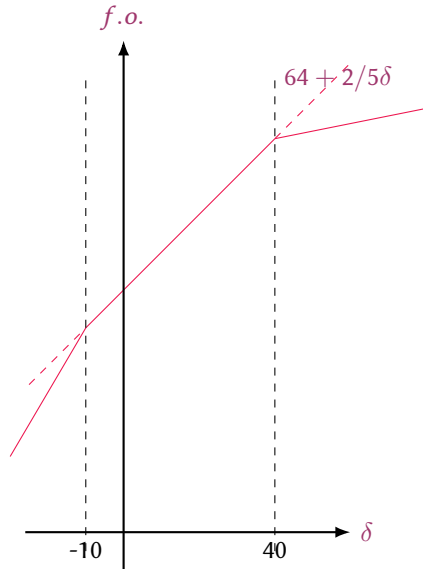
I: $1/5(60 + \delta) - 1/4 \cdot 40 + 0 \cdot 0 = 12 + \delta/5 - 10 = 2 + \delta/5$

II: $-1/5(60 + \delta) + 1/2 \cdot 40 + 0 \cdot 0 = -60/5 + 20 - \delta/5 = 8 - 1/5\delta$

Risk that RHS becomes negative

Eg: if $\delta = -10 \implies$ tableau stays optimal but not feasible $\implies$ apply dual simplex

Graphical Representation



(III) Add a variable

$$\begin{aligned} \max \quad & 5x_0 + 6x_1 + 8x_2 \\ & 6x_0 + 5x_1 + 10x_2 \leq 60 \\ & 8x_0 + 4x_1 + 4x_2 \leq 40 \\ & x_0, x_1, x_2 \geq 0 \end{aligned}$$

Reduced cost of x_0 ? $c_j + \sum \pi_i a_{ij} = +1 \cdot 5 - \frac{2}{5} \cdot 6 + (-1)8 = -\frac{27}{5}$

To make worth entering in basis:

- increase its cost
- decrease the technological coefficient in constraint II: $5 - 2/5 \cdot 6 - a_{20} > 0$

(IV) Add a constraint

$$\begin{aligned}\max \quad & 6x_1 + 8x_2 \\ & 5x_1 + 10x_2 \leq 60 \\ & 4x_1 + 4x_2 \leq 40 \\ & 5x_1 + 6x_2 \leq 50 \\ & x_1, x_2 \geq 0\end{aligned}$$

Final tableau not in canonical form, need to iterate with dual simplex

	x_1	x_2	x_3	x_4	x_5	$-z$	b
x_2	0	1	$1/5$	$-1/4$		0	2
x_1	1	0	$-1/5$	$1/2$		0	8
	0	0	$-1/5$	-1	1	0	-2
	0	0	$-2/5$	-1	0	1	-64

(V) change in a technological coefficient:

	x_1	x_2	x_3	x_4	$-z$	b
x_3	5	$10 + \delta$	1	0	0	60
x_4	4	4	0	1	0	40
	6	8	0	0	1	0

- first effect on its column
- then look at c
- finally look at b

	x_1	x_2	x_3	x_4	$-z$	b
x_2	0	$(10 + \delta)1/5 + 4(-1/4)$	$1/5$	$-1/4$	0	2
x_1	1	$(10 + \delta)(-1/5) + 4(1/2)$	$-1/5$	$1/2$	0	8
	0	$-2/5\delta$	$-2/5$	-1	1	-64

Relevance of Sensitivity Analysis

- The dominant application of LP is mixed integer linear programming.
- In this context it is extremely important being able to begin with a model instantiated in one form followed by a sequence of problem modifications
 - row and column additions and deletions,
 - variable fixingsinterspersed with resolves

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